Applied Statistics - Exercise 8

1. Mean and median of two datasets (Theory)

Consider two datasets $x_1, \ldots, x_n$ and $y_1, \ldots, y_m$. Note that they have different lengths. Let $\bar{x}$ be the sample mean of the first, and $\bar{y}$ the sample mean of the second. Consider the combined dataset $x_1, \ldots, x_n, y_1, \ldots, y_m$ with $m+n$ elements, obtained by concatenating the two original datasets.

a. Is it true that the sample mean of the combined dataset is equal to $\frac{\bar{x}+\bar{y}}{2}$? If yes, provide a proof, if no, provide a counterexample.

No, that is not the case. For instance;

```
x <- rnorm(100)
y <- rnorm(80)
(mean(x) + mean(y))/2
```

```
## [1] -0.0748811
```

```
mean(c(x,y))
```

```
## [1] -0.07844319
```

b. Consider the case where $m = n$, i.e. the two datasets have the same size. In this special case, is the sample mean of the combined dataset equal to $\frac{\bar{x}+\bar{y}}{2}$? If yes, provide a proof, if no, provide a counterexample.

```
x <- rnorm(100)
y <- rnorm(100)
(mean(x) + mean(y))/2
```

```
## [1] 0.08953897
```

```
mean(c(x,y))
```

```
## [1] 0.08953897
```

$$
\begin{aligned}
\bar{x} + \bar{y} &= \frac{x_1 + \ldots + x_n}{n} + \frac{y_1 + \ldots + y_n}{n} \\
&= \frac{x_1 + \ldots + x_n + y_1 + \ldots + y_n}{n} \\
\frac{\bar{x} + \bar{y}}{2} &= \frac{x_1 + \ldots + x_n + y_1 + \ldots + y_n}{2n}
\end{aligned}
$$

By definition the mean of the combined dataset of x and y, as it has 2n samples.

c. Consider now the sample medians Med_x and Med_x of the two datasets, in the general case of $m \neq n$. Is it true that the sample median of the combined dataset is equal to $\frac{Med_x + Med_y}{2}$? If yes, provide a proof, if no, provide a counterexample.

No, that is not the case.

```r
x <- rnorm(100)
y <- rnorm(70)
(median(x) + median(y))/2
```

```
## [1] 0.1439919
```

```r
median(c(x,y))
```

```
## [1] 0.1026767
```

d. In the special case of $m = n$, is the sample median of the combined dataset is equal to $\frac{Med_x + Med_y}{2}$? If yes, provide a proof, if no, provide a counterexample.

Also not true in the case that $m = n$.

```r
x <- rnorm(100)
y <- rnorm(100)
(median(x) + median(y))/2
```

```
## [1] 0.057769
```

```r
median(c(x,y))
```

```
## [1] 0.03037502
```

2. Recognizing plots (Theory)

Consider the following distributions:

- $N(0, 1)$
- $N(0, 8)$
- $N(5, 2)$
- $Exp(2)$
- $Exp(1/2)$

The following plots report histograms, kernel density estimates, and empirical distribution functions, each for a different dataset of 150 points generated from the above distributions. For each plot, say which type of plot it is (i.e. if it's a histogram, a kernel density estimate or an empirical distribution function), and identify from which of the above distributions it was generated.

Datasets:

- Dataset 1: Empirical distribution function, generated by $N(0, 1)$
- Dataset 2: Kernel density estimate, generated by $N(0, 1)$
- Dataset 3: Histogram, generated by $Exp(2)$
- Dataset 4: Kernel density estimate, generated by $Exp(1/2)$
- Dataset 5: Histogram, generated by $N(0, 8)$
- Dataset 6: Histogram, generated by $N(5, 2)$
- Dataset 7: Kernel density estimate, generated by $Exp(1/2)$
- Dataset 8: Empirical distribution function, generated by $Exp(1/2)$
- Dataset 9: Kernel density estimate, generated by $N(5, 2)$
- Dataset 10: Empirical distribution function, generated by $N(5, 2)$
- Dataset 11: Empirical distribution function, generated by $Exp(2)$
- Dataset 12: Histogram, generated by $Exp(2)$

3. Chopsticks factory (Theory)

You are running a chopstick factory: due to the production process, the chopsticks are not of the exactly same length. As part of quality control you want to check that all chopsticks have approximately the same length. Suppose you produce 2000 chopsticks each day:

a. Choose an appropriate model for the chopsticks length. *Hint*: consider the model usually used to account for random variations in productions, experimental measures, etc.

The appropriate model for the chopstick length would be a normal distribution $X \sim N(\mu, \sigma^2)$

b. Let x_i be the length of the i-th chopstick produced today. At the end of the day you see that $\sum_i x_i = 45999$ and $\sum_i x_i^2 = 1058019$. Estimate μ and σ^2 for the distribution you chose in point a. *Hint*: look closely at how the variance is estimated and rework the formula so to be able to use the available data.

$$
\begin{aligned}
\bar{x} &= \frac{45999}{2000} \approx 229.995 \\
S_n^2 &= \frac{1}{n-1} \sum_{i=1}^{n} x_i^2 - 2x, \bar{x} + \bar{x}^2 \\
&= \frac{1}{n-1} \left(\sum_{i=1}^{n} (x_i^2) - 2\bar{x} \sum_{i=1}^{n} (x_i) + n\bar{x}^2 \right) \\
&= \frac{1}{1999} (1058019 - 2 \cdot 22.99 \cdot 45999 + 2000 \cdot 22.99^2) \\
&= \frac{64.99}{1999} \approx 0.0326
\end{aligned}
$$

c. Give an estimate of the probability that the length of a random chopstick is within the interval $[22.5, 23.5]$.

```
pnorm(23.5, 22.99, sqrt(0.0326)) - pnorm(22.5, 22.99, sqrt(0.0326))
```

```
## [1] 0.994308
```

4. Simple Statistics (R)

Consider the `firstchi` dataset, available in the `UsingR` package, which you can load using the `library(UsingR)` statement. Using R functions, compute the following numerical statistics for the dataset.

- the sample mean
- the sample variance
- the 30th empirical percentile
- the median
- the MAD
- the IQR

You can refer to Section 2.3 of *Using R for introductory statistics*.

```
mean(firstchi)
```

```
## [1] 23.97701
```

```
var(firstchi)
```

```
## [1] 39.11574
```

```
quantile(firstchi, 0.3)
```

```
## 30%
##  21
```

```r
median(firstchi)
```

```
## [1] 23
```

```r
mad(firstchi)
```

```
## [1] 4.4478
```

```r
IQR(firstchi)
```

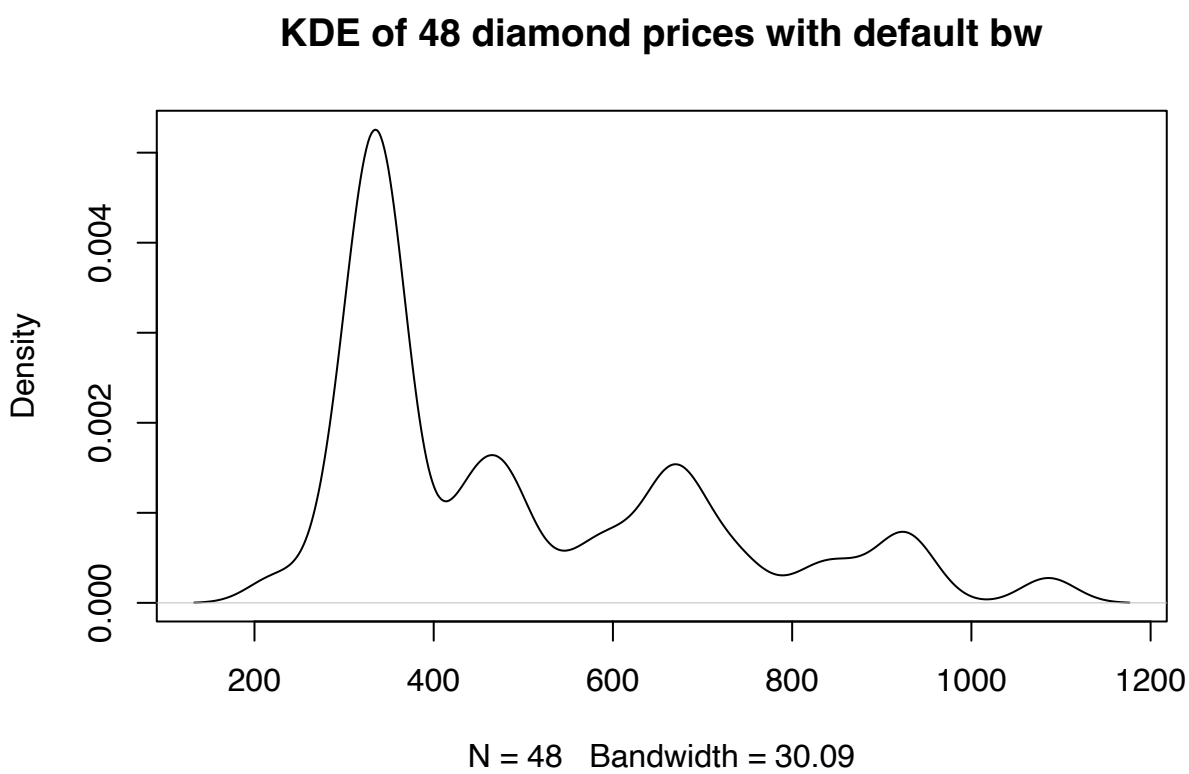
```
## [1] 6
```

5. Plotting distributions (R)

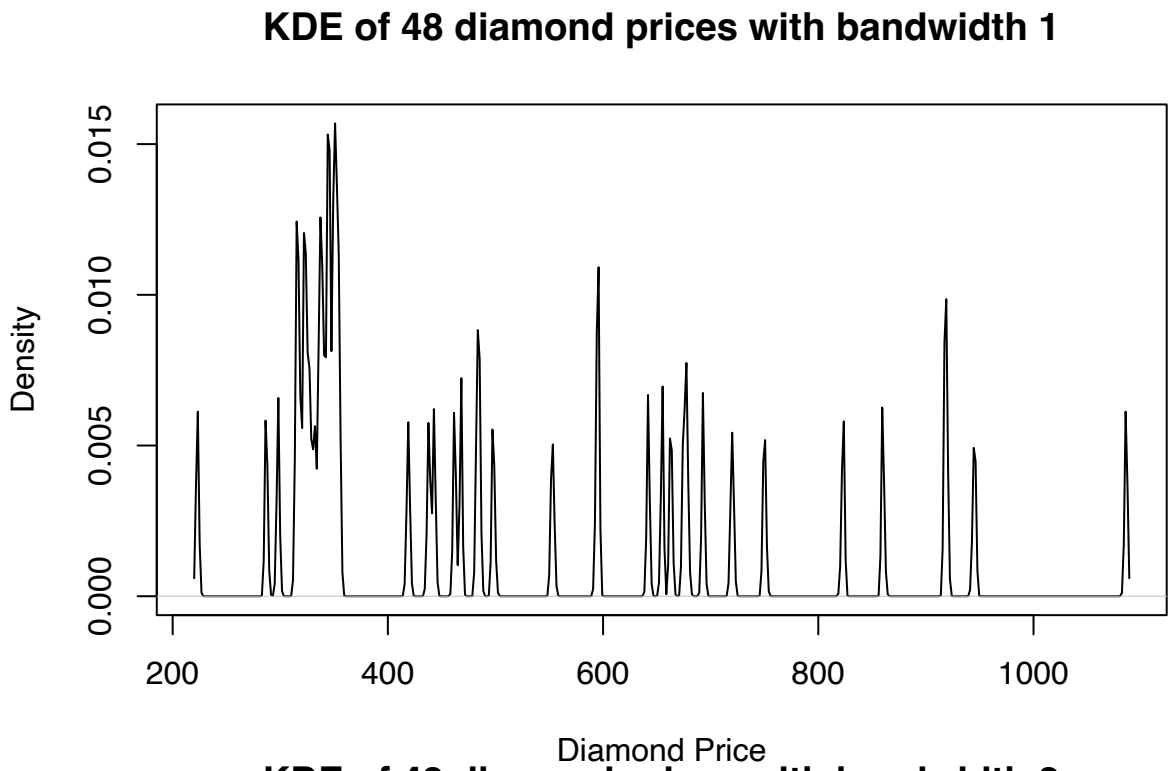
The `diamond` dataset of the `UsingR` package contains the price in Singapore dollars of 48 diamond rings, along with their size in carats.

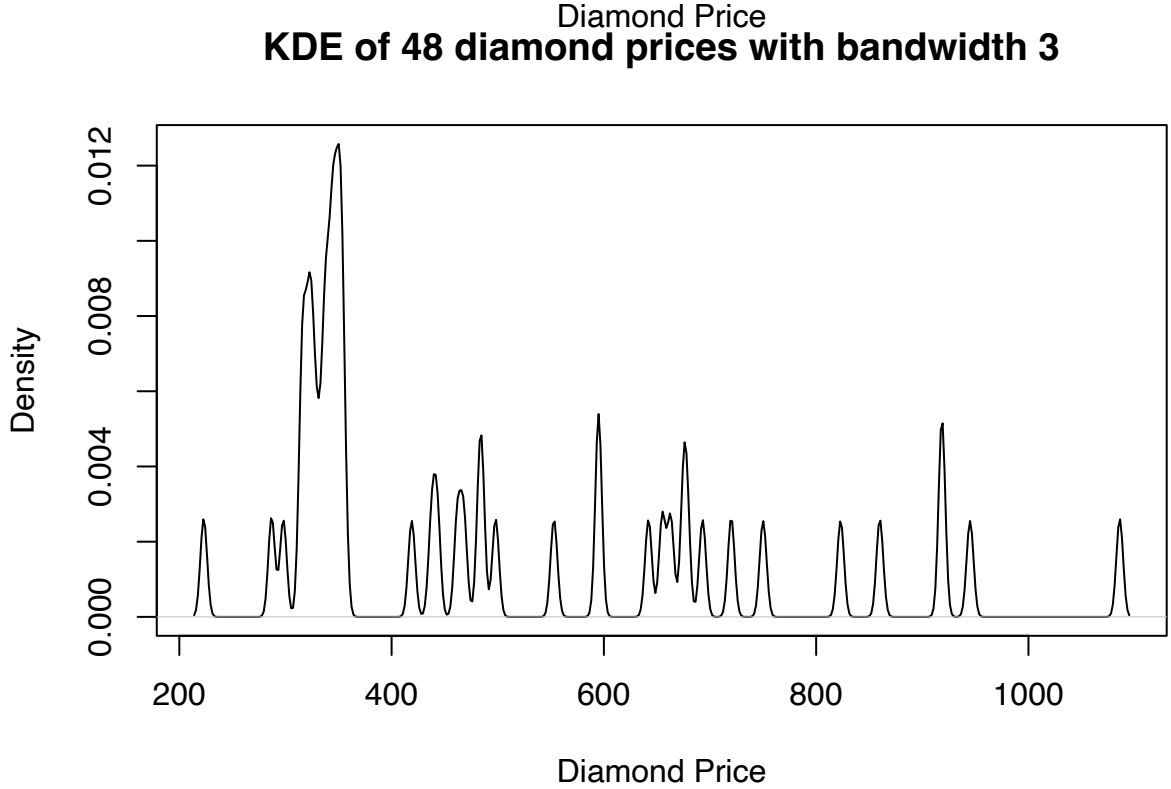
1. Plot the kernel density estimate of prices. Try different bandwidths. How many modes are there? Look also at the empirical cumulative distribution function. Discuss your findings.

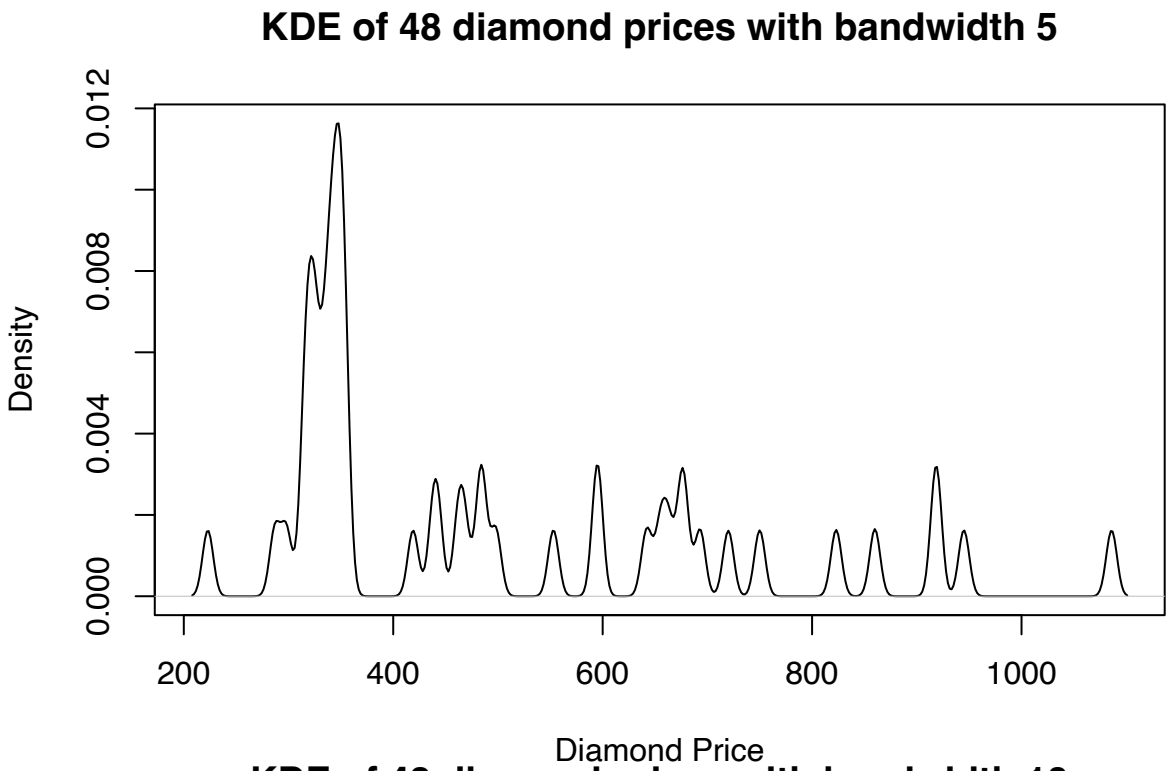
```r
dplot <- function (bw) {
  plot(density(diamond$price, bw=bw),
       main=sprintf("KDE of 48 diamond prices with bandwidth %d", bw),
       xlab="Diamond Price")
}
plot(density(diamond$price, bw="SJ"),
     main="KDE of 48 diamond prices with default bw")
```

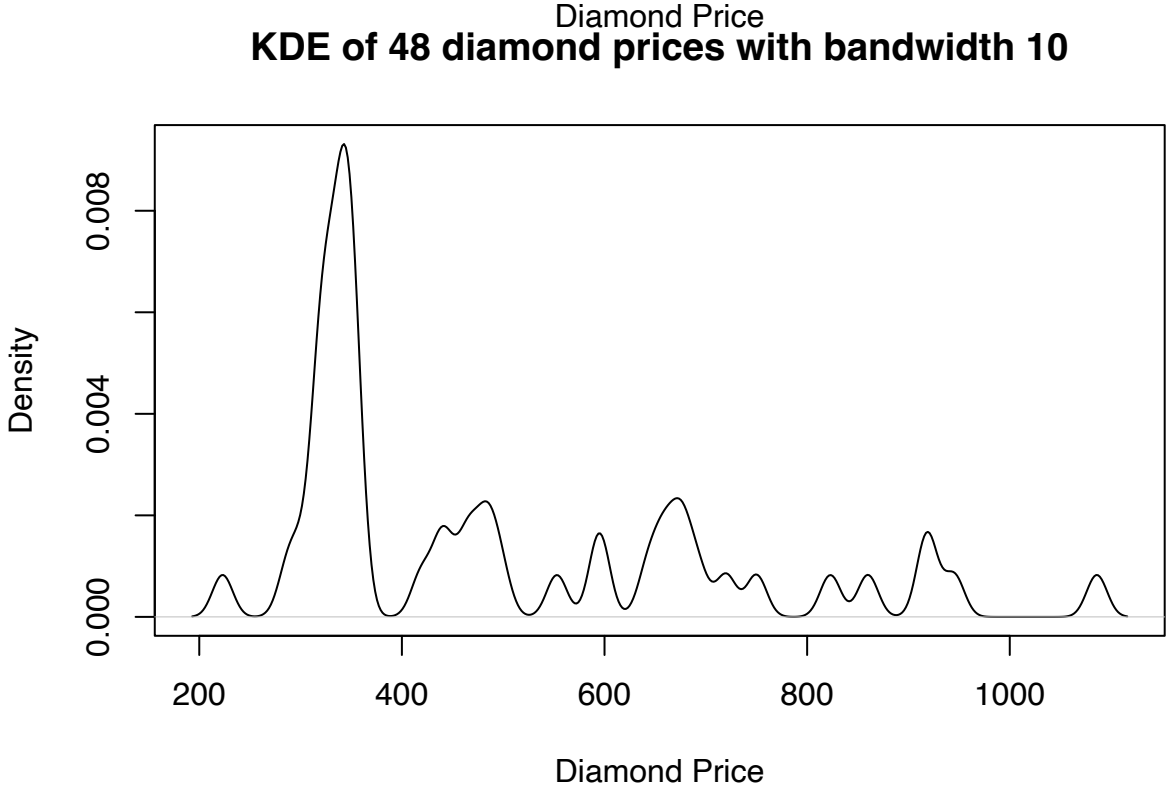


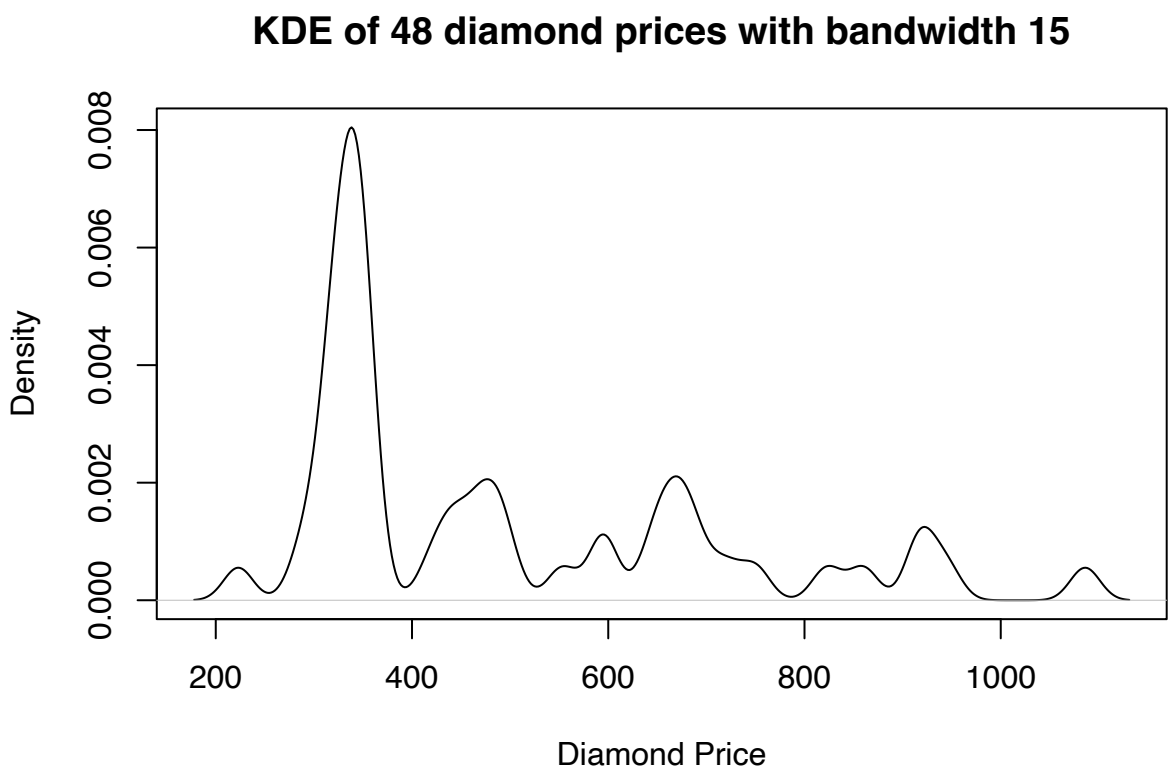

```r
sapply(c(1, 3, 5, 10, 15), dplot)
```


KDE of 48 diamond prices with bandwidth 1
Density
0.000
0.005
0.010
0.015
200
400
600
800
1000
Diamond Price

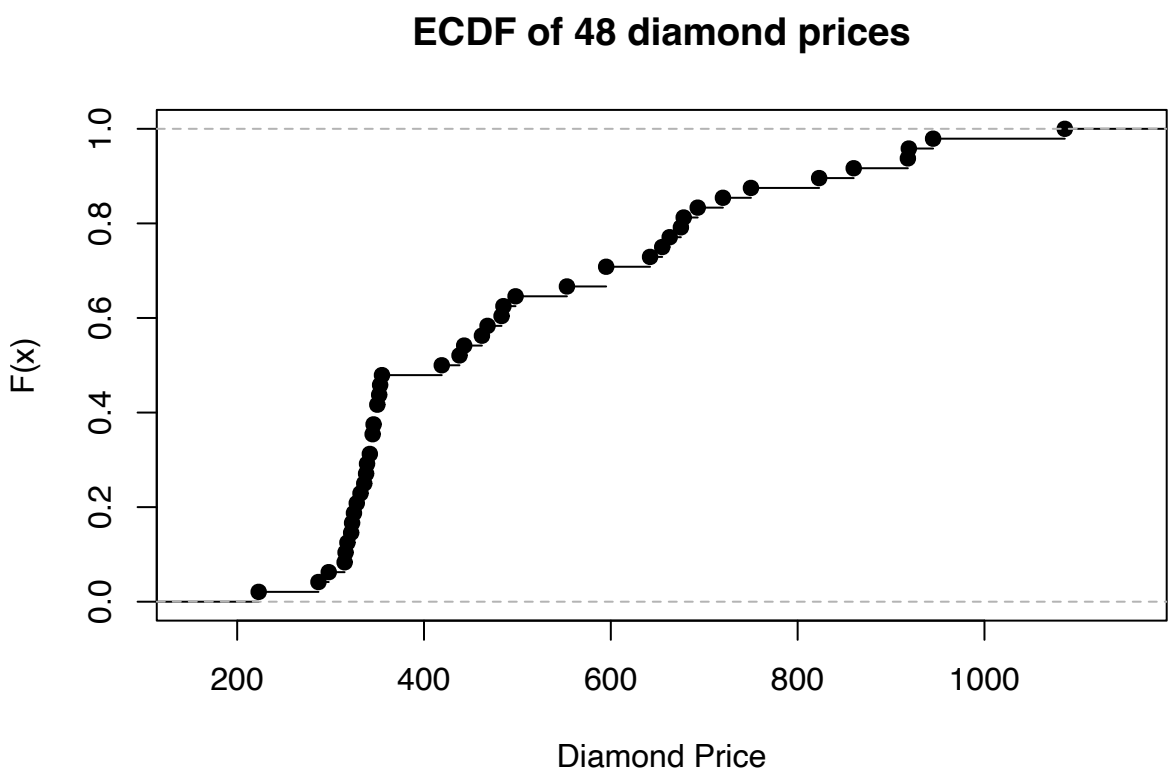

KDE of 48 diamond prices with bandwidth 3
Density
0.000
0.004
0.008
0.012
200
400
600
800
1000
Diamond Price


KDE of 48 diamond prices with bandwidth 5
Density
0.000
0.004
0.008
0.012
200
400
600
800
1000
Diamond Price


KDE of 48 diamond prices with bandwidth 10
Density
0.000
0.004
0.008
200
400
600
800
1000
Diamond Price



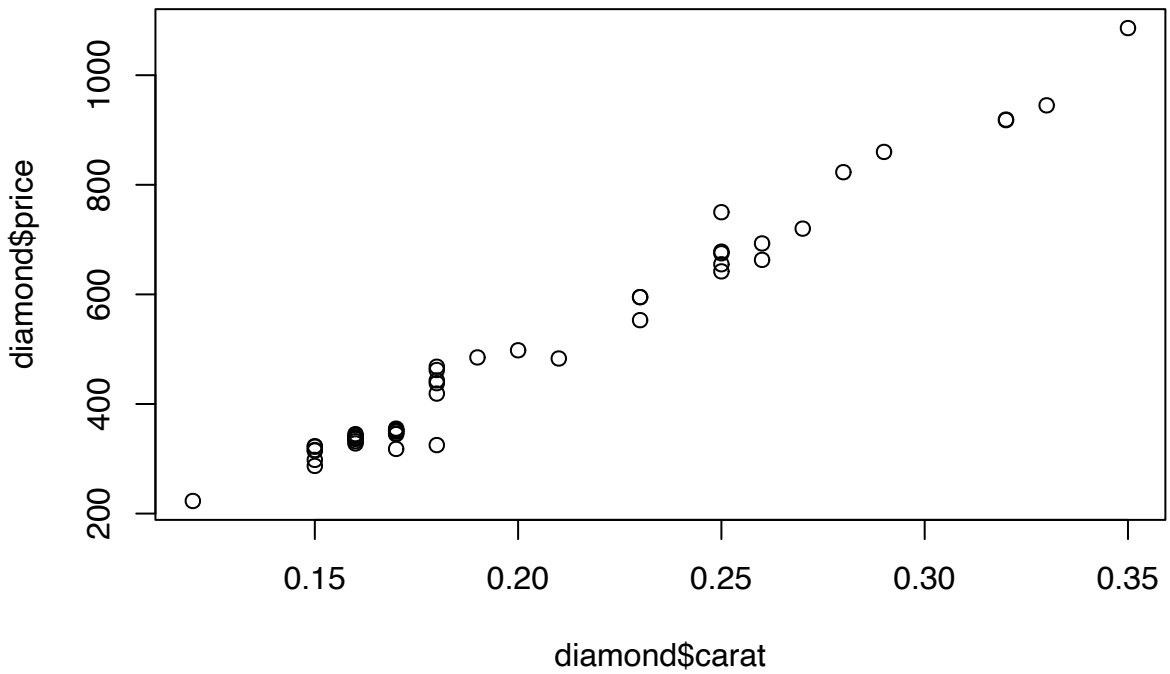

```
plot(ecdf(diamond$price),
     main="ECDF of 48 diamond prices",
     xlab="Diamond Price",
     ylab="F(x)")
```



There seem to be one main mode and around 3 less significant modes.

2. Plot a scatterplot of prices versus sizes. Does any relation between the two quantities show up?

```
plot(diamond$carat, diamond$price)
```



It would seem that the price goes up as the size goes up.

```
cor(diamond$carat, diamond$price)
```

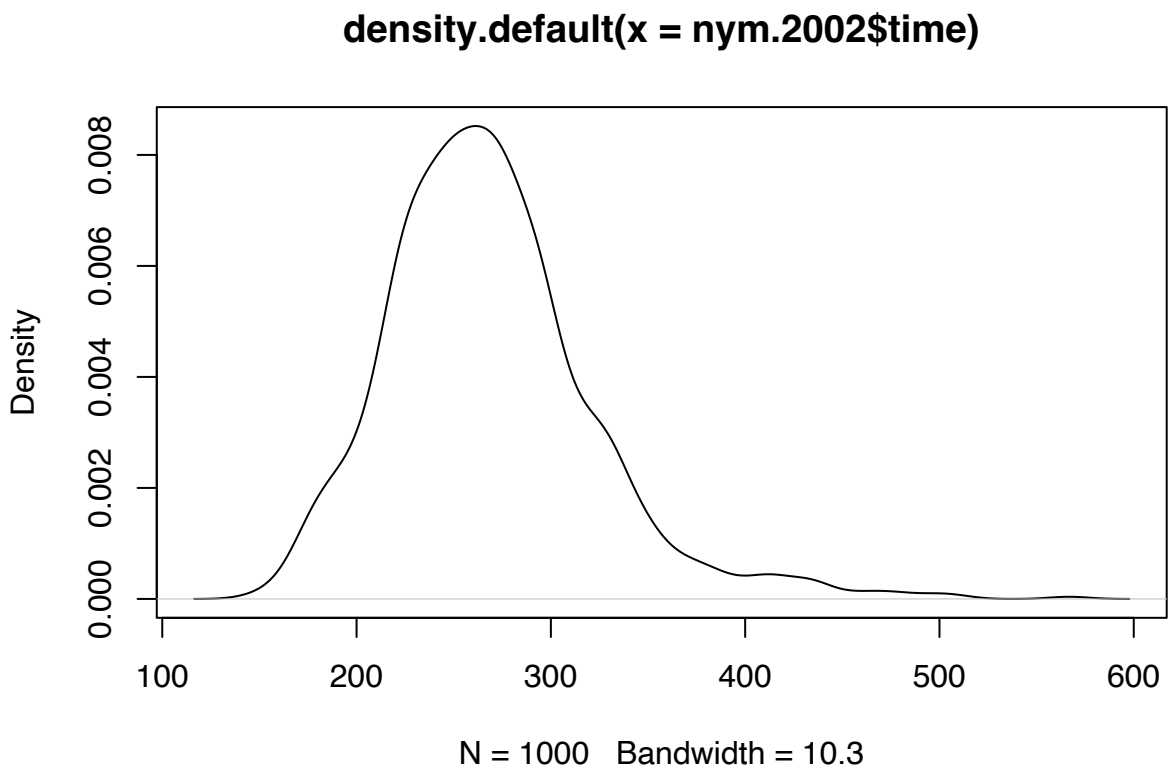
```
## [1] 0.9890707
```

6. New York marathon (R)

The `nym.2002` dataset (in the UnsingR package) contains information about the times taken by participants of the 2002 New York marathon, along with information like age and gender. First of all, bring the dataset into scope by loading the UsingR library: `library(UsingR)`.

a. Plot the kernel density estimate of the `time` column. Given that we have other information available in the dataset, is such a histogram informative? Discuss about this.

```
plot(density(nym.2002$time))
```



It is useful to get an idea of how long it typically took people to complete the marathon. But it does not tell us anything about which groups took longer or shorter, it is very general.

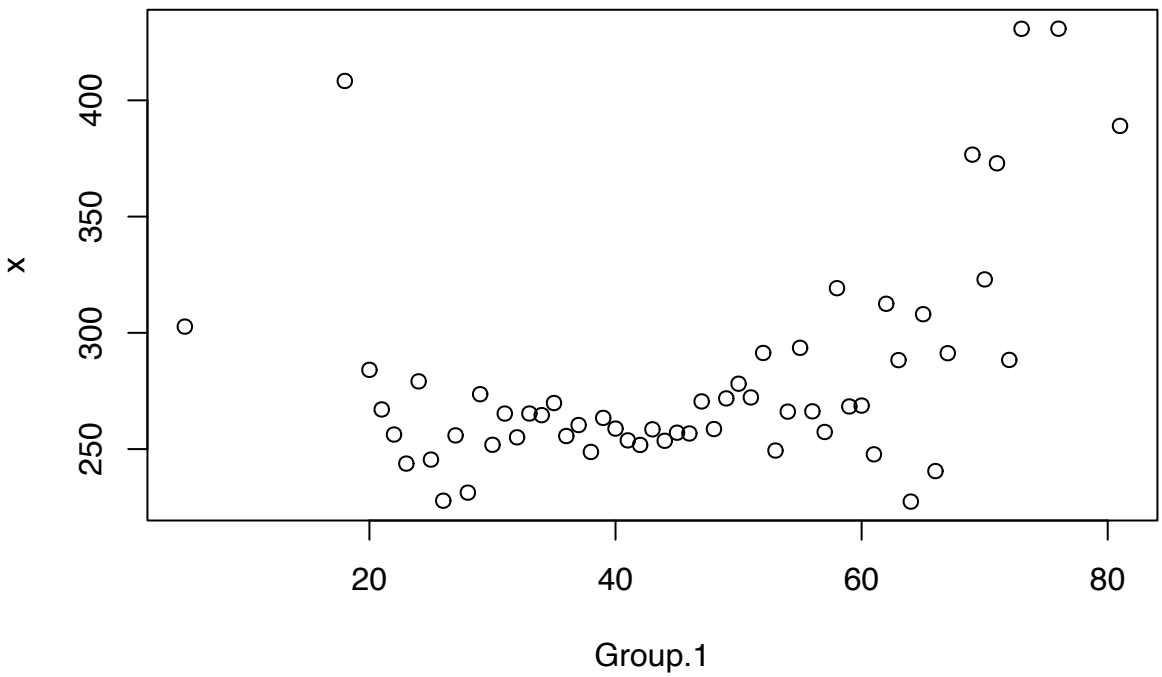
b. Consider the variable age in combination with `time`. Compute the median time for each age group. To this end, you can use the `aggregate` function, which takes two vectors of the same length (the first one are the values, the other the grouping variables) and a function to aggregate the values belonging to the same group. Build the following plot: on the x axis we have the age, and on the y axis we have the corresponding median time. What do you observe?

```
aggregate(nym.2002$time, by=list(nym.2002$age), FUN=median)
```

```
##    Group.1        x
## 1        5 302.6833
## 2       18 408.3333
## 3       20 284.0833
## 4       21 267.0500
## 5       22 256.2667
## 6       23 243.7667
## 7       24 279.1250
## 8       25 245.4833
## 9       26 227.8000
## 10      27 255.8833
## 11      28 231.2750
## 12      29 273.6083
## 13      30 251.8500
## 14      31 265.2833
## 15      32 255.0667
## 16      33 265.3000
## 17      34 264.6000
## 18      35 269.8167
## 19      36 255.6167
```

```
## 20       37 260.3500
## 21       38 248.7750
## 22       39 263.4167
## 23       40 258.8167
## 24       41 253.7500
## 25       42 251.7833
## 26       43 258.5000
## 27       44 253.5667
## 28       45 257.0500
## 29       46 256.7167
## 30       47 270.5000
## 31       48 258.6167
## 32       49 271.8000
## 33       50 278.1167
## 34       51 272.2167
## 35       52 291.3417
## 36       53 249.3833
## 37       54 266.1083
## 38       55 293.5583
## 39       56 266.2333
## 40       57 257.3583
## 41       58 319.2083
## 42       59 268.3333
## 43       60 268.7083
## 44       61 247.7500
## 45       62 312.5000
## 46       63 288.2667
## 47       64 227.4167
## 48       65 308.0167
## 49       66 240.5500
## 50       67 291.2417
## 51       69 376.6500
## 52       70 322.9833
## 53       71 372.9167
## 54       72 288.3500
## 55       73 430.7583
## 56       76 430.8167
## 57       81 389.0000
```

```
plot(aggregate(nym.2002$time, by=list(nym.2002$age), FUN=median))
```

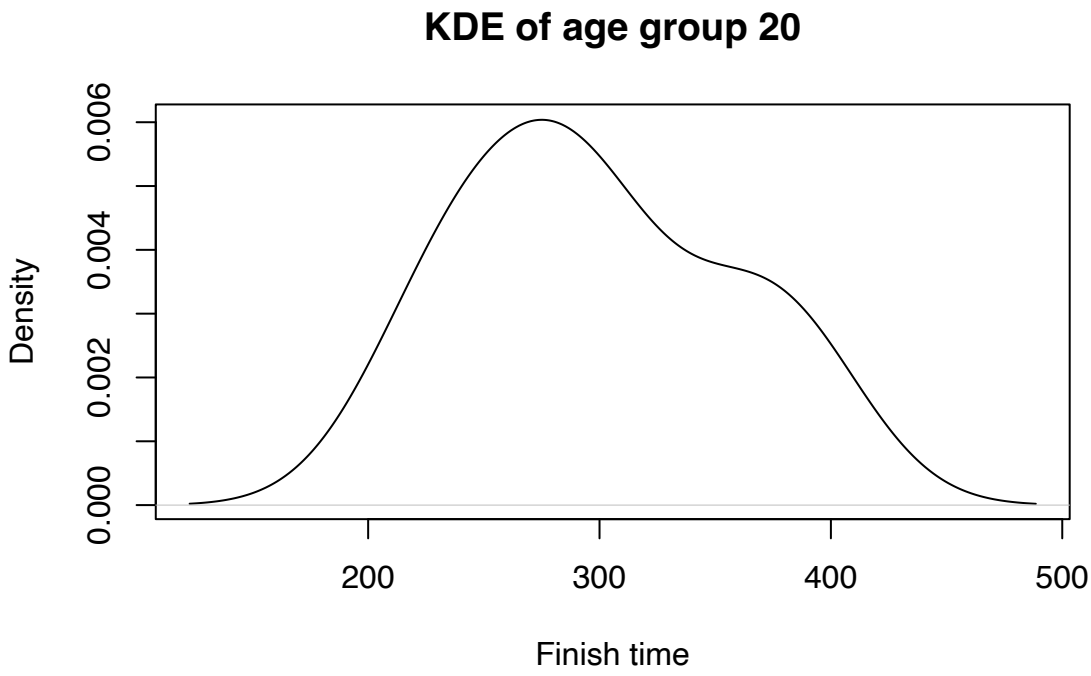


There does not seem to be a linear correlation between age and finish time. However, past the age of ~55 there is a decline in finishing time that could look linear.

c. Plot the kernel density estimate for each age group. To do so, you can use the following tools:

1. Get the set of ages in the dataset using the `unique` function on the `age` column of the dataframe
2. Select the rows corresponding to an age using the `subset` function, which is described on page 159 of Verzani's book. In short, you can use `subset(dataframe, subset = column_name == value)` to select all the rows with the given `value` in the column `column_name`.

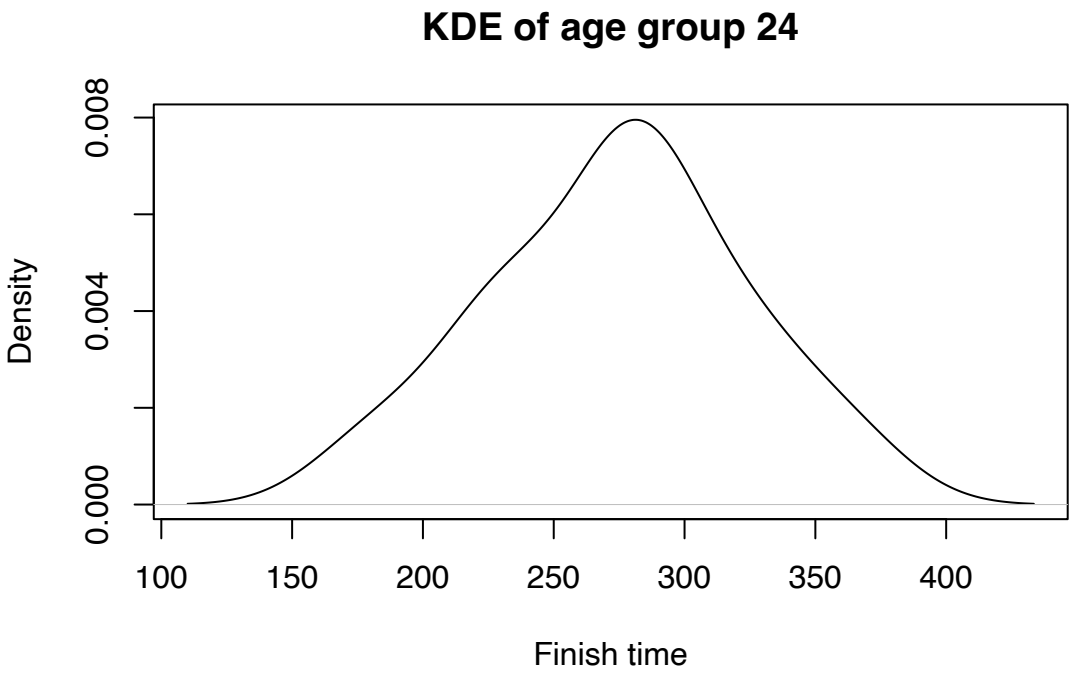
```
plot.kde.age <- function (n) {
  df <- subset(nym.2002, age == n)
  if (dim(df)[1] > 1) {
    plot(density(df$time),
         main=sprintf("KDE of age group %d", n),
         xlab="Finish time")
  }
}
unique(nym.2002$age) %>% sort() %>%
sapply(., plot.kde.age)
```

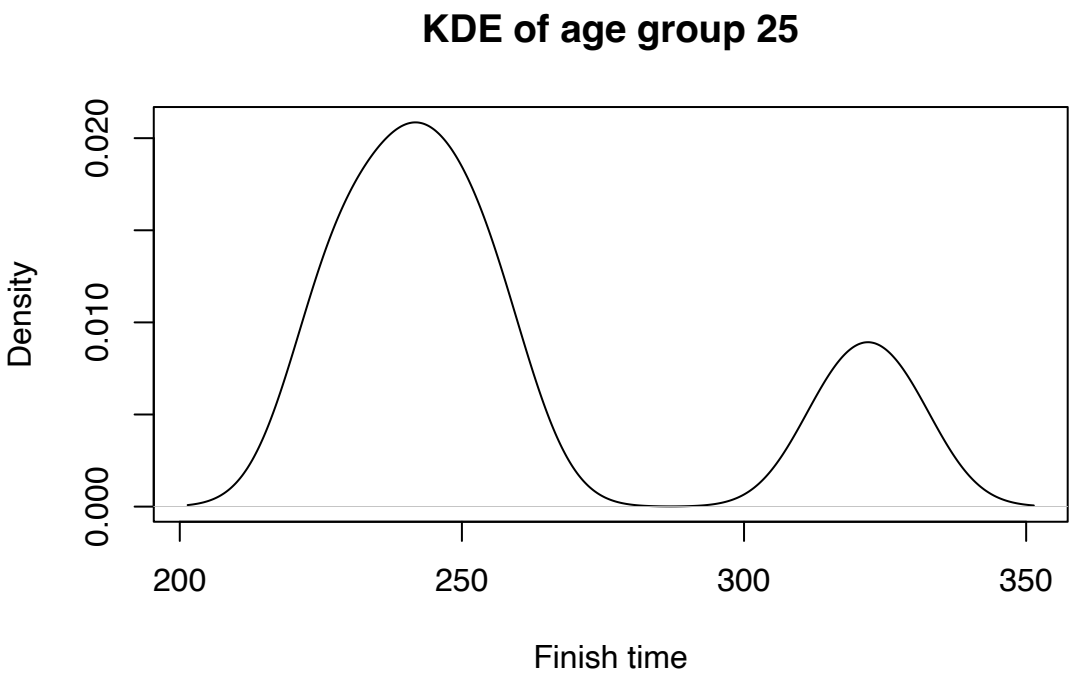

KDE of age group 20
Density
0.000
0.002
0.004
0.006
200
300
400
500
Finish time

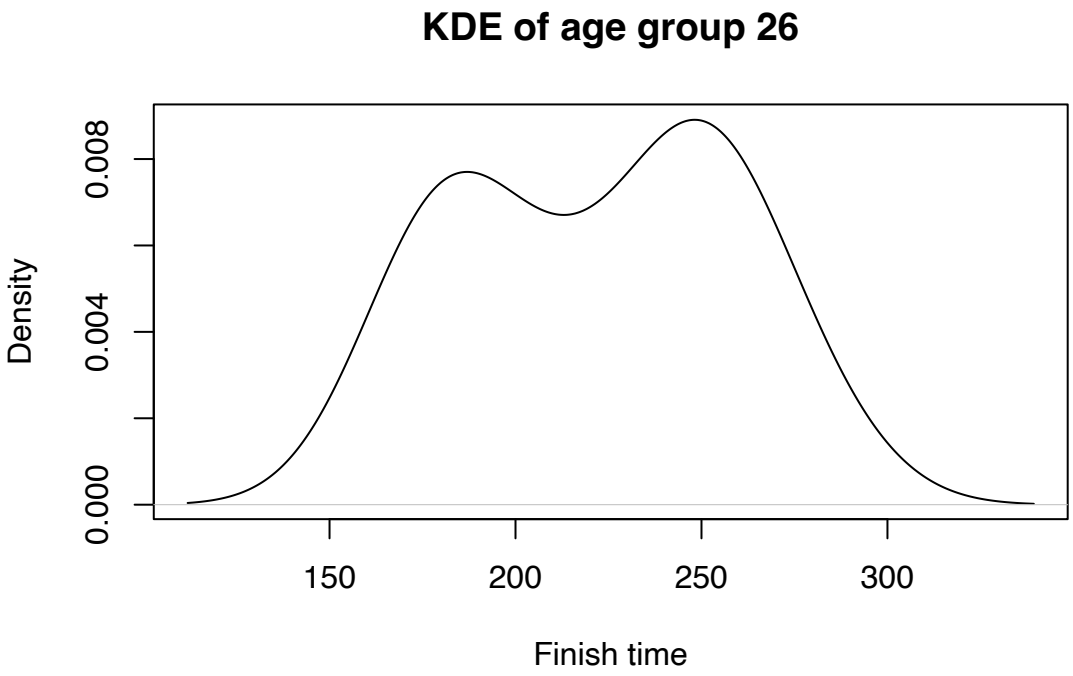
KDE of age group 21
Density
0.000
0.004
0.008
150
200
250
300
350
Finish time

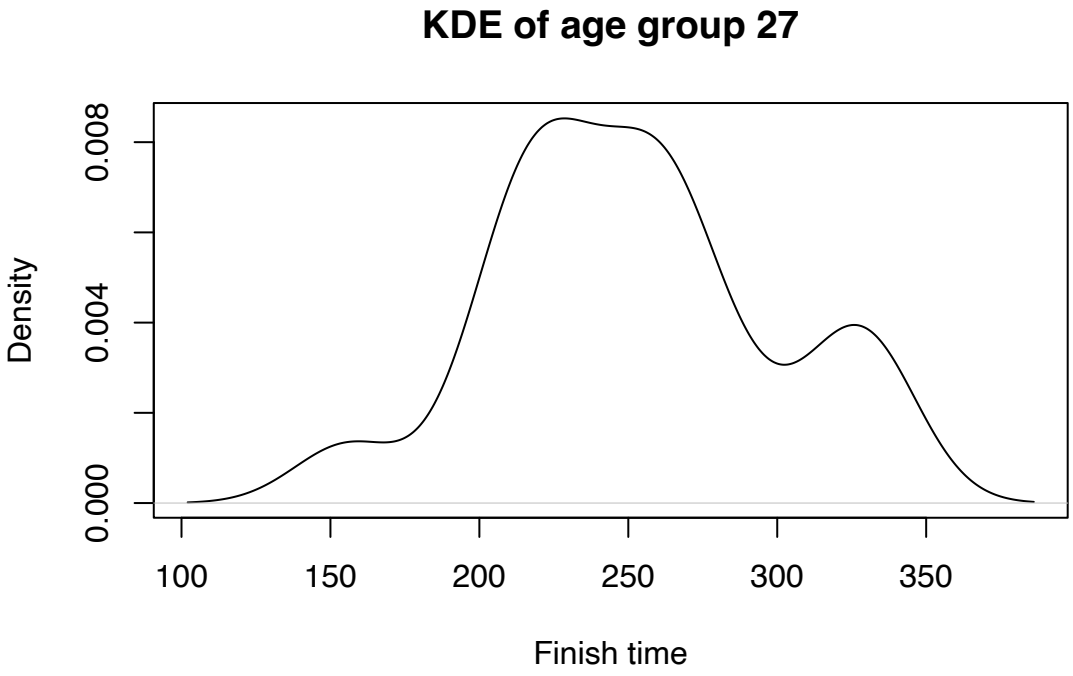
KDE of age group 22
Density
0.000
0.004
0.008
150
200
250
300
350
400
Finish time

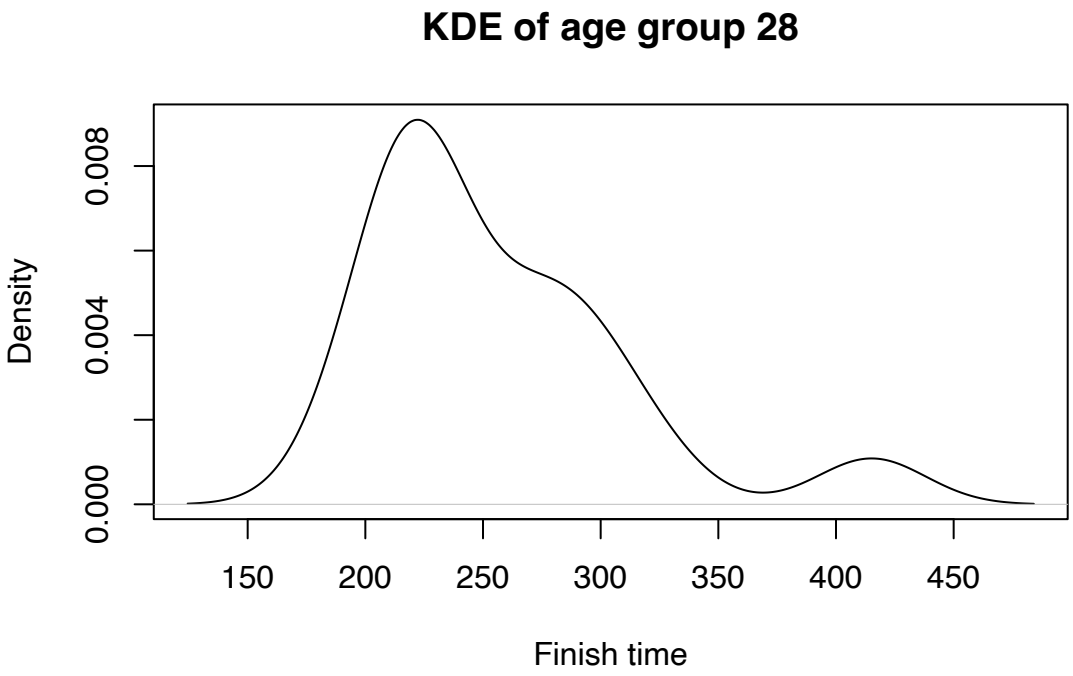
KDE of age group 23
Density
0.000
0.004
0.008
150
200
250
300
350
400
Finish time

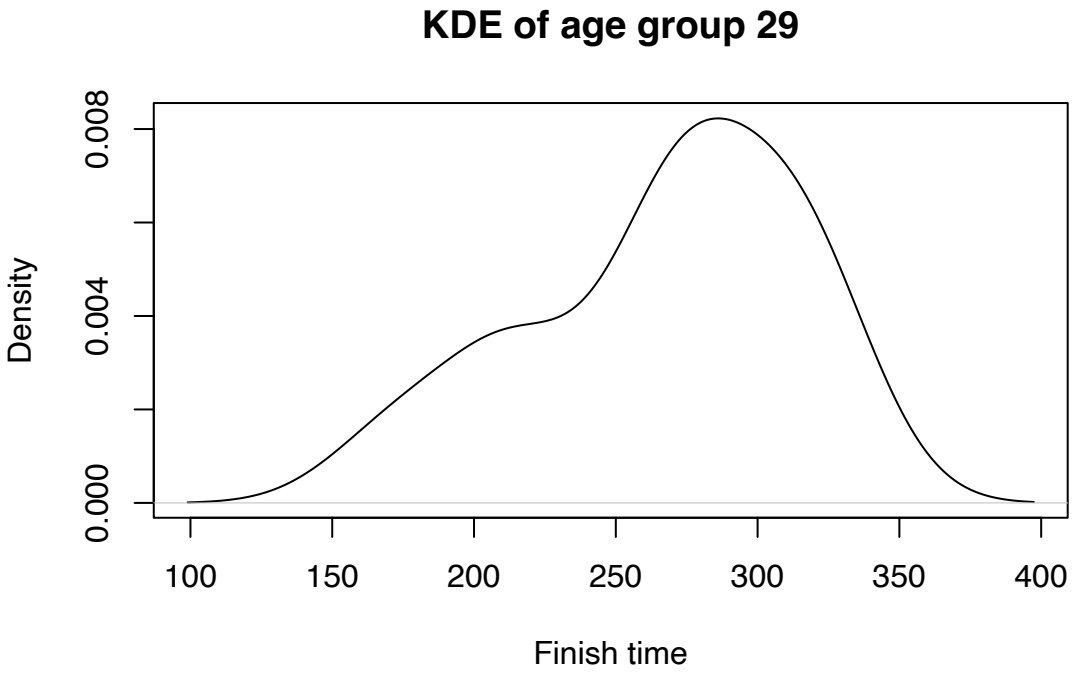

KDE of age group 24
Density
0.000
0.004
0.008
100
150
200
250
300
350
400
Finish time

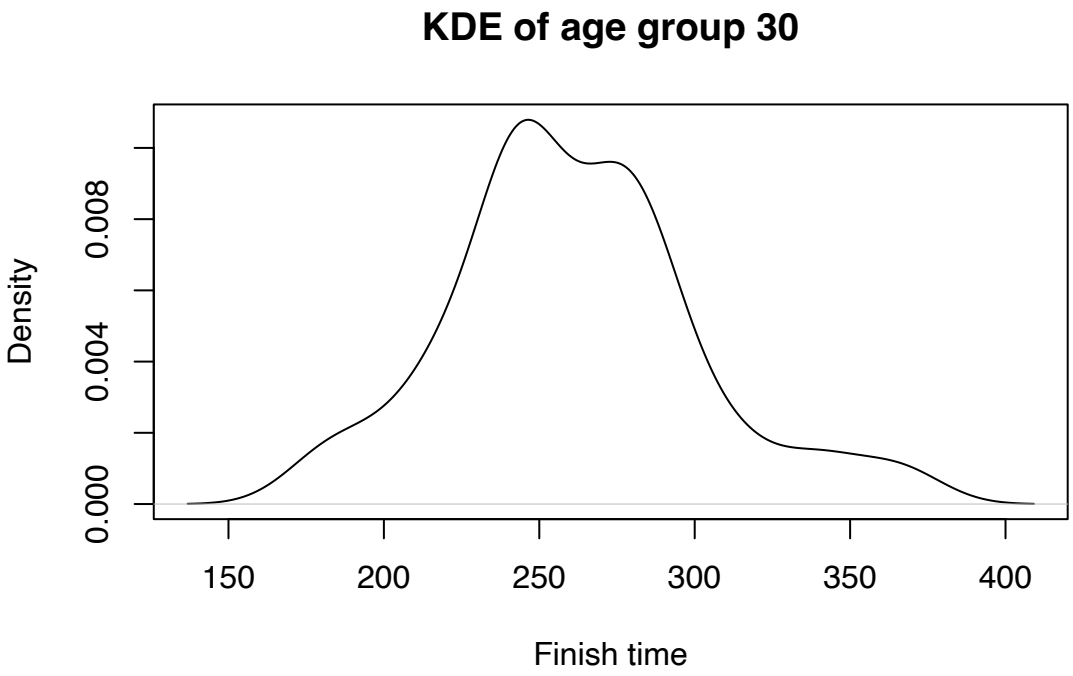

KDE of age group 25
Density
0.000
0.010
0.020
200
250
300
350
Finish time

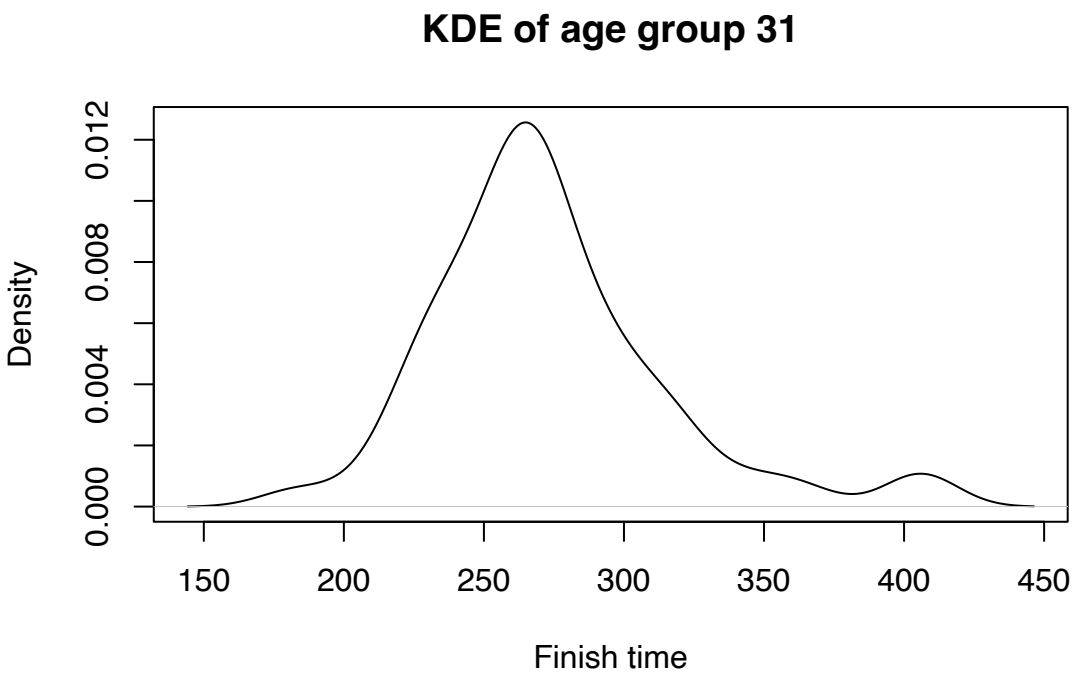

KDE of age group 26
Density
0.000
0.004
0.008
150
200
250
300
Finish time

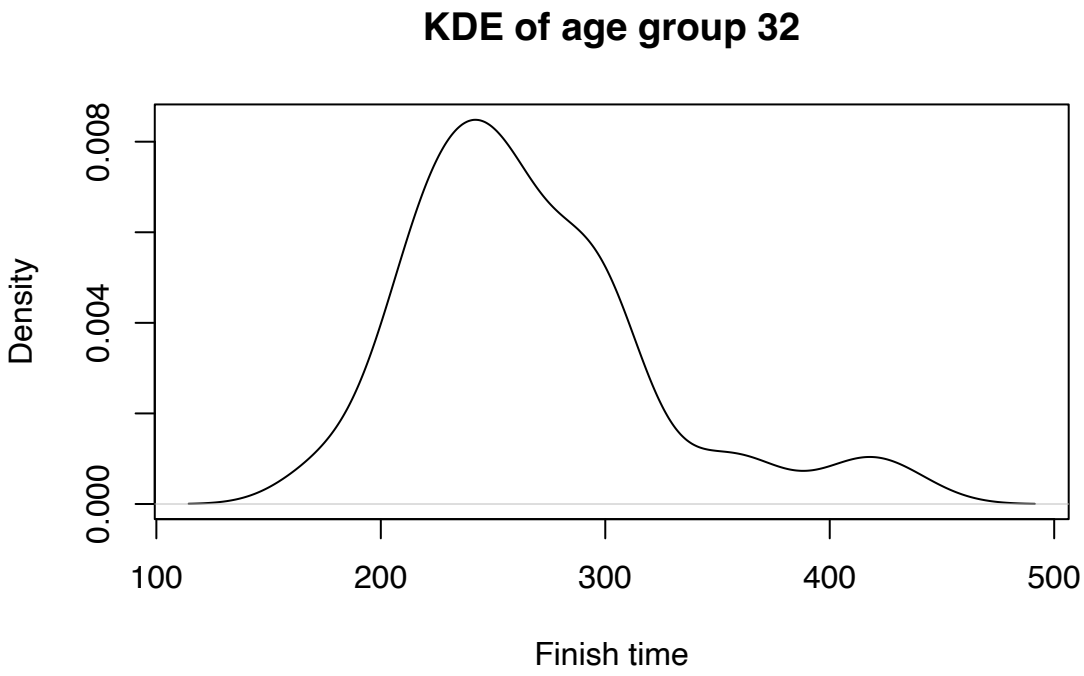

KDE of age group 27
Density
0.000
0.004
0.008
100
150
200
250
300
350
Finish time

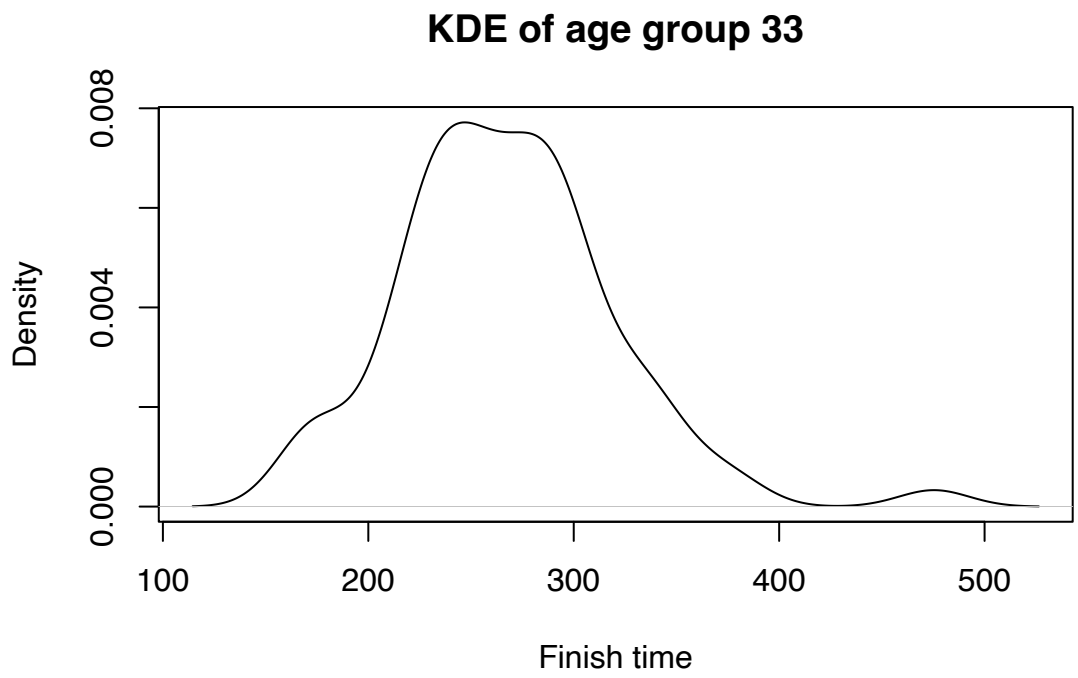

KDE of age group 28
Density
0.000
0.004
0.008
150
200
250
300
350
400
450
Finish time

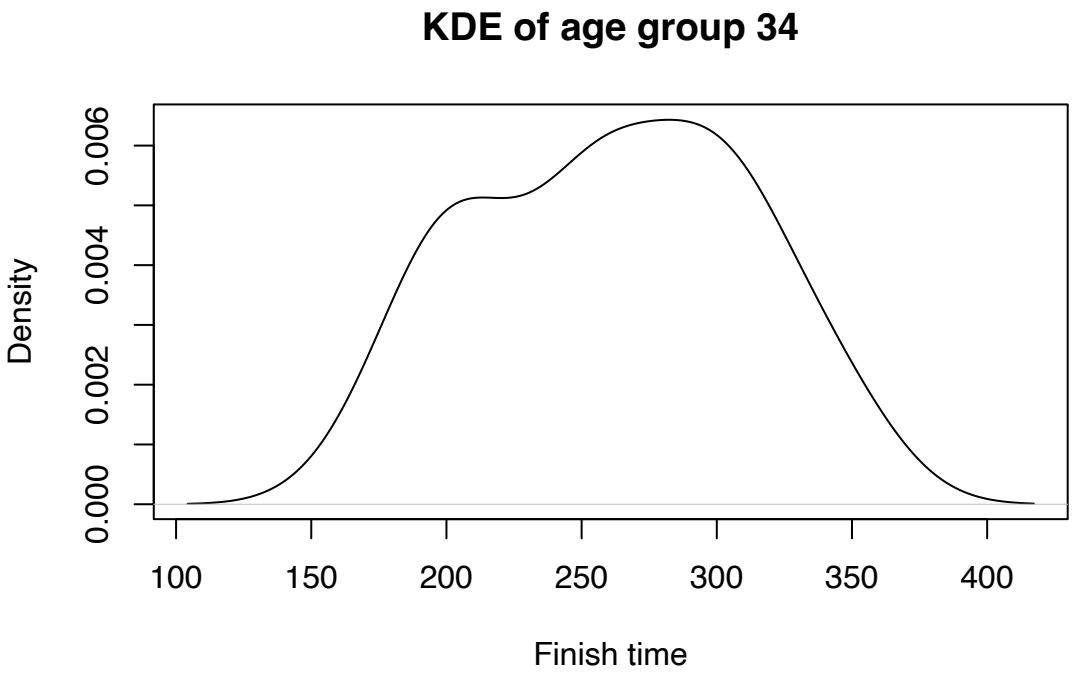

KDE of age group 29
Density
0.000
0.004
0.008
100
150
200
250
300
350
400
Finish time

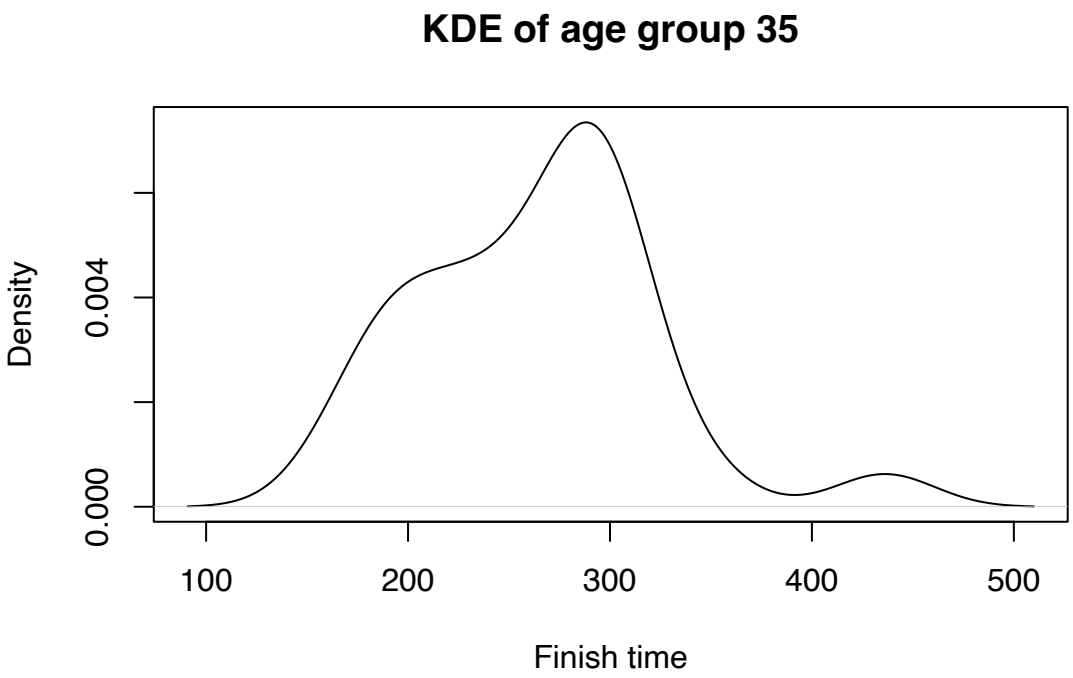

KDE of age group 30
Density
0.000
0.004
0.008
150
200
250
300
350
400
Finish time

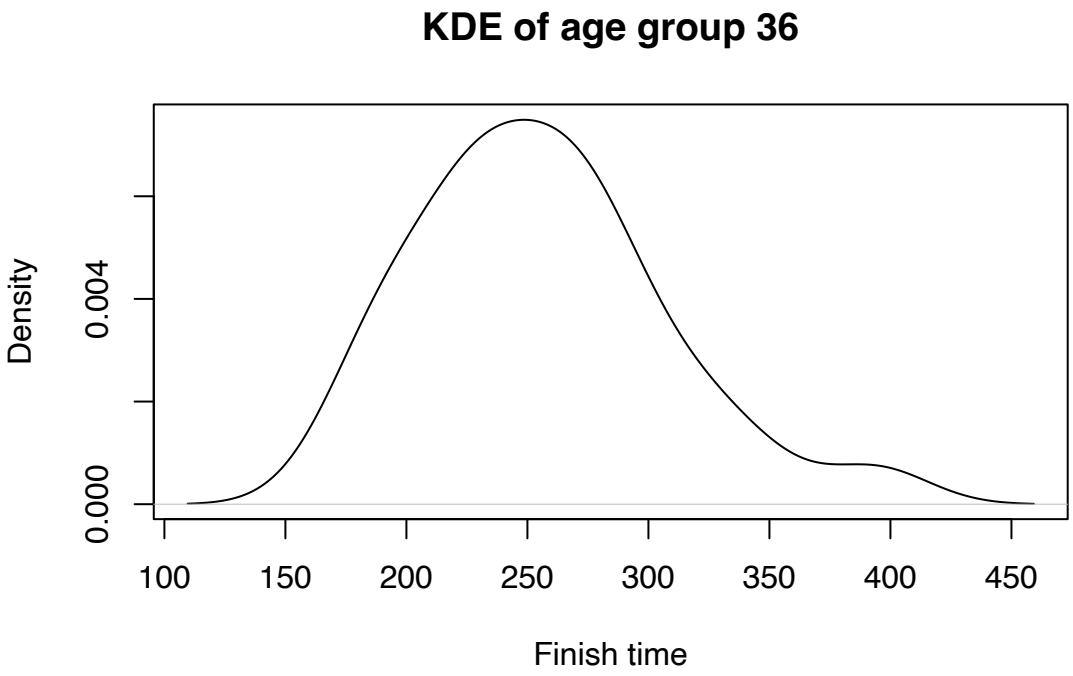

KDE of age group 31
Density
0.000
0.004
0.008
0.012
150
200
250
300
350
400
450
Finish time


KDE of age group 32
Density
0.000
0.004
0.008
100
200
300
400
500
Finish time


KDE of age group 33
Density
0.000
0.004
0.008
100
200
300
400
500
Finish time


KDE of age group 34
Density
0.000
0.002
0.004
0.006
100
150
200
250
300
350
400
Finish time


KDE of age group 35
Density
0.000
0.004
100
200
300
400
500
Finish time


KDE of age group 36
Density
0.000
0.004
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150
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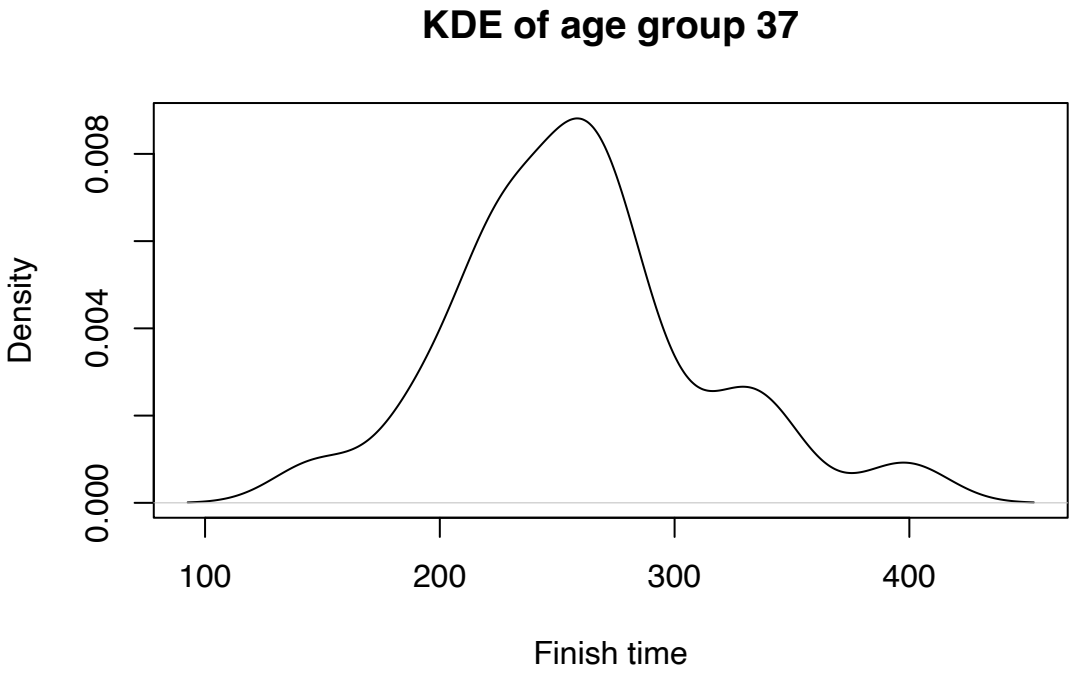
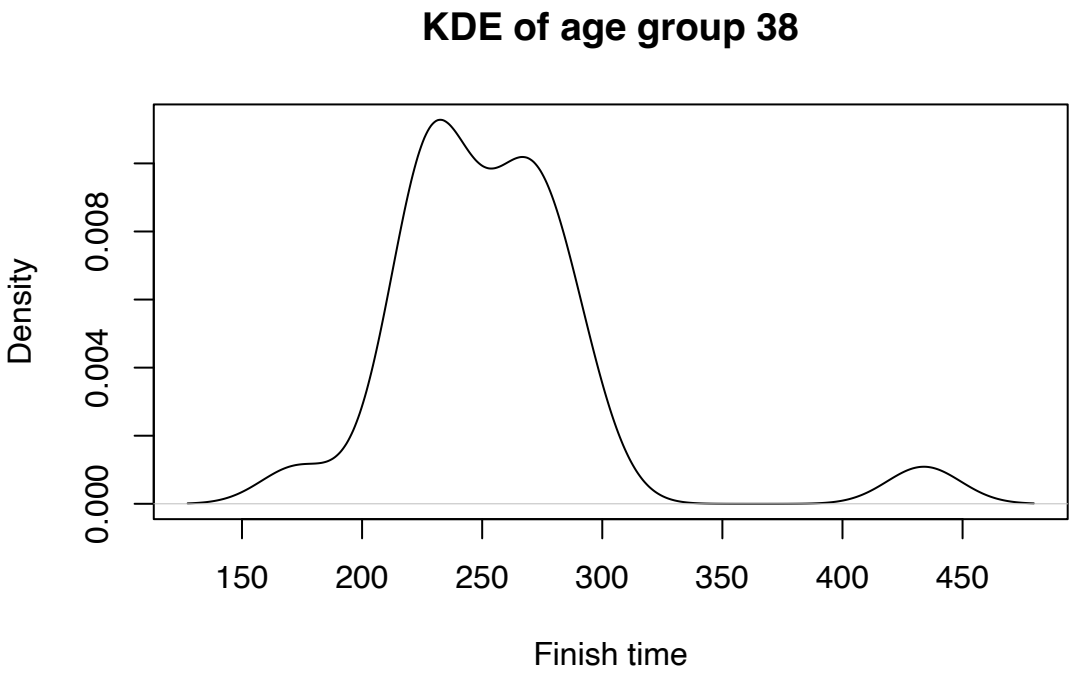
300
350
400
450
Finish time


KDE of age group 37
Density
0.000
0.004
0.008
100
200
300
400
Finish time


KDE of age group 38
Density
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0.004
0.008
150
200
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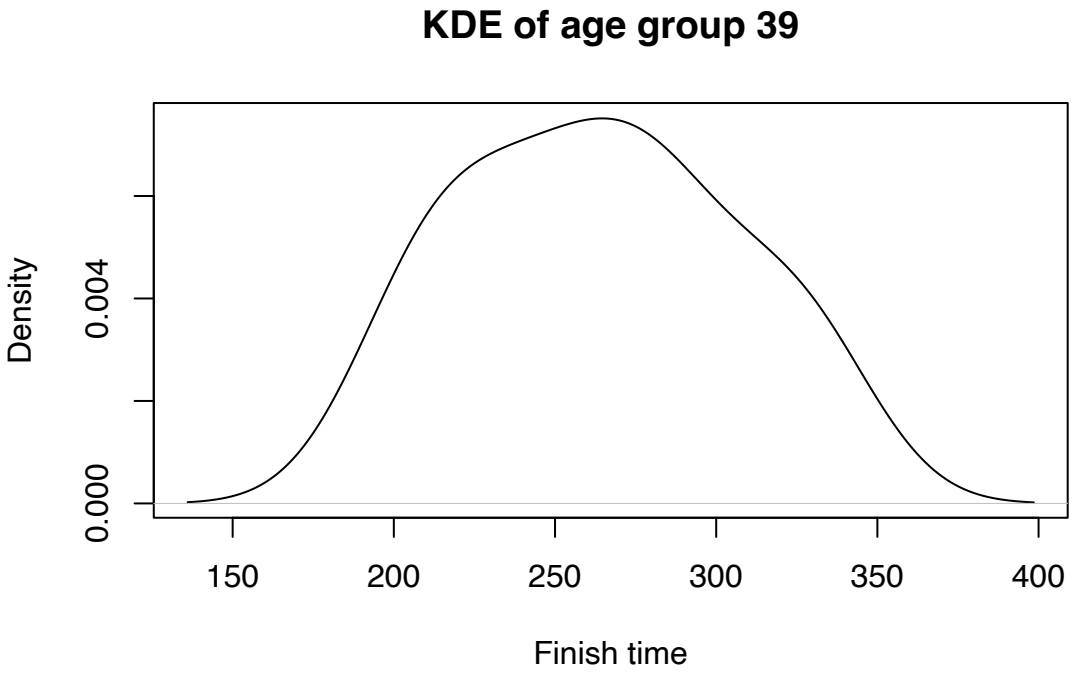
300
350
400
450
Finish time


KDE of age group 39
Density
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200
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300
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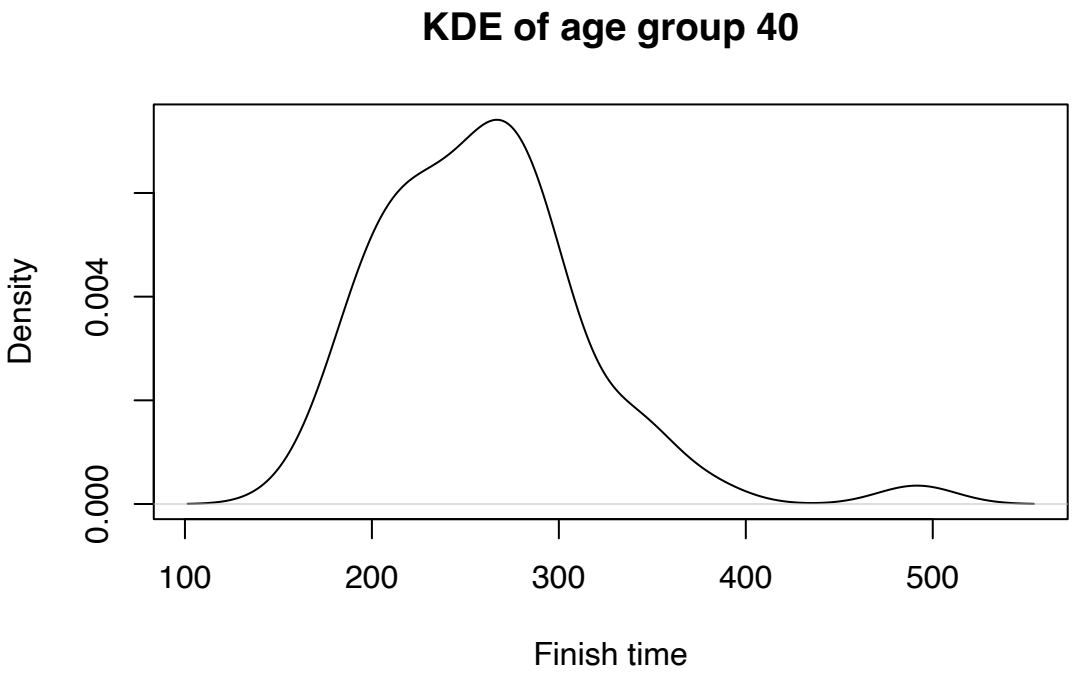
400
Finish time


KDE of age group 40
Density
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200
300
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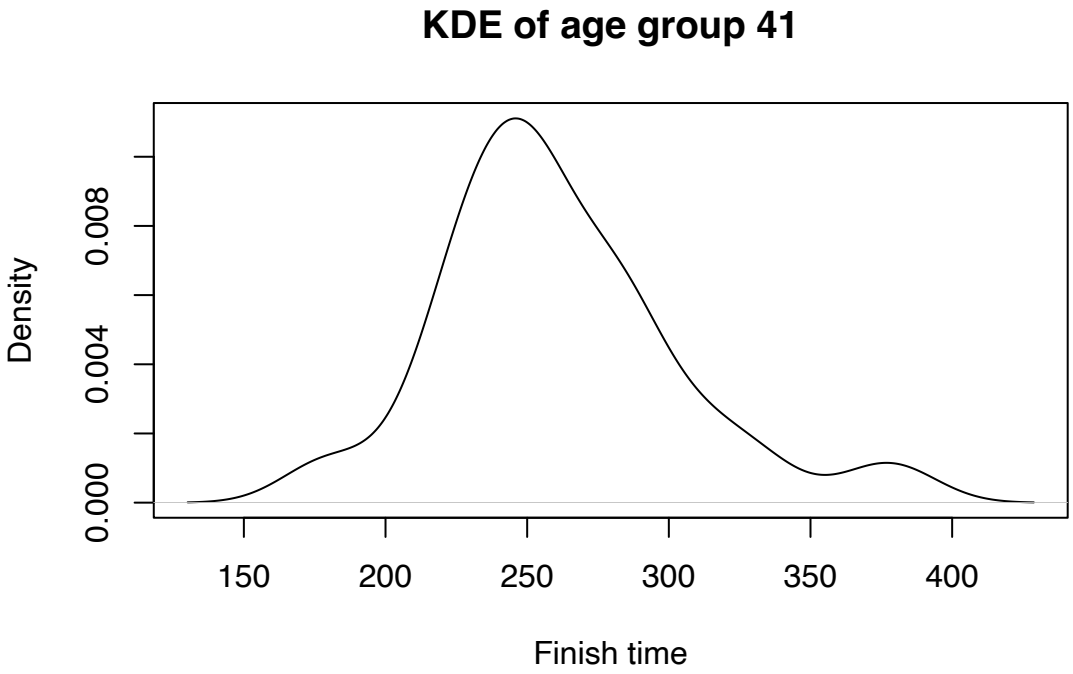
500
Finish time


KDE of age group 41
Density
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0.004
0.008
150
200
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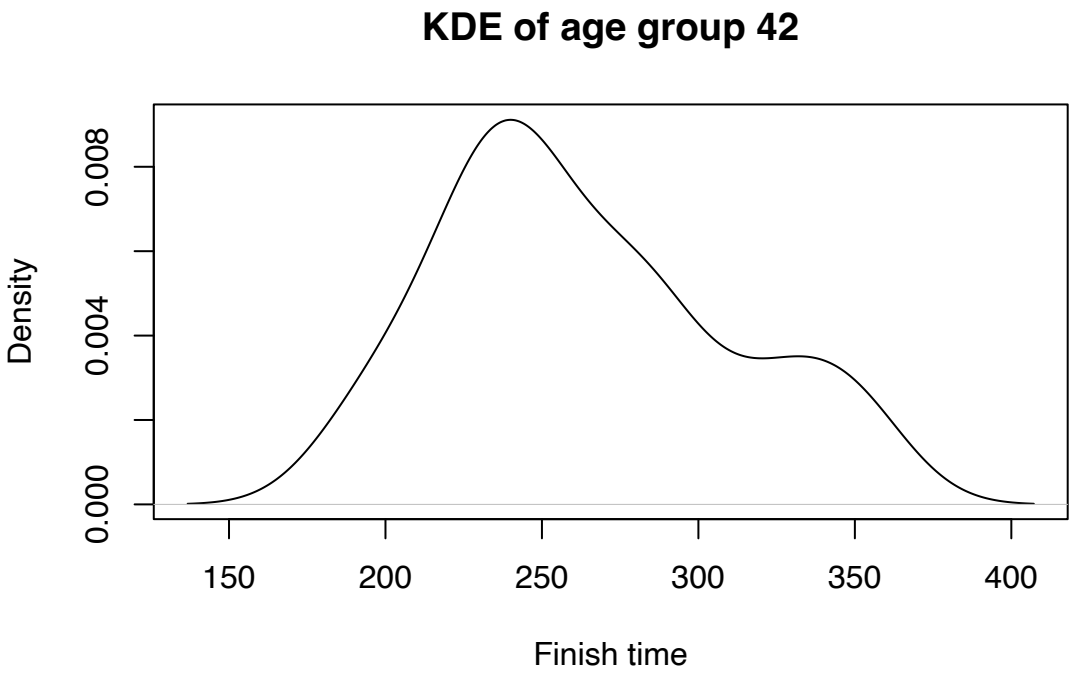
300
350
400
Finish time


KDE of age group 42
Density
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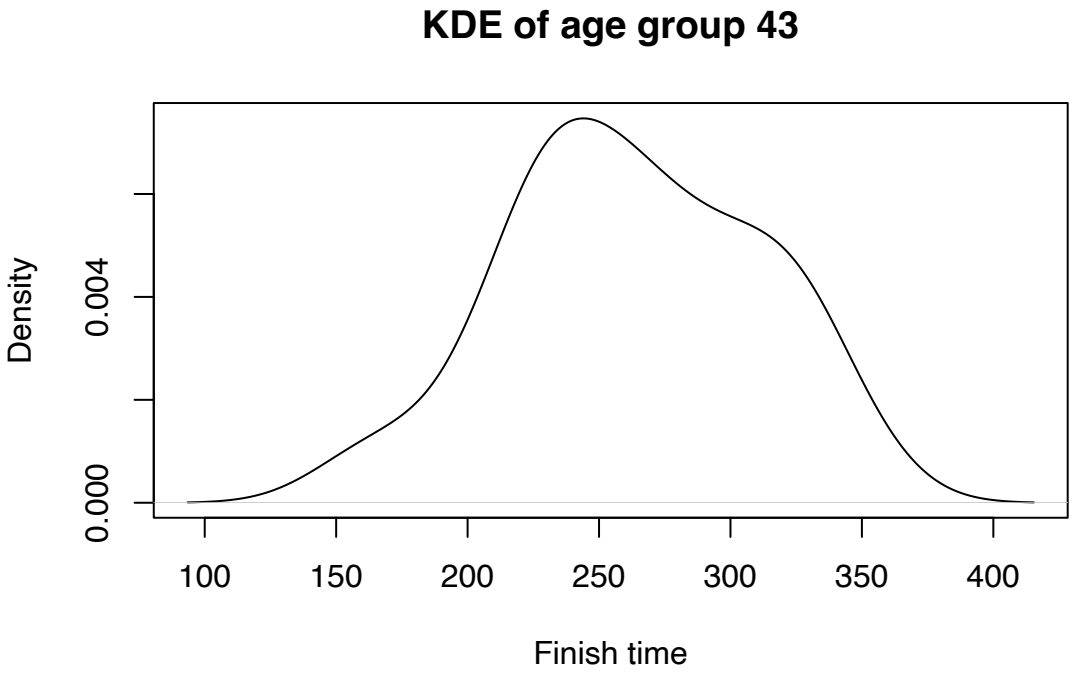
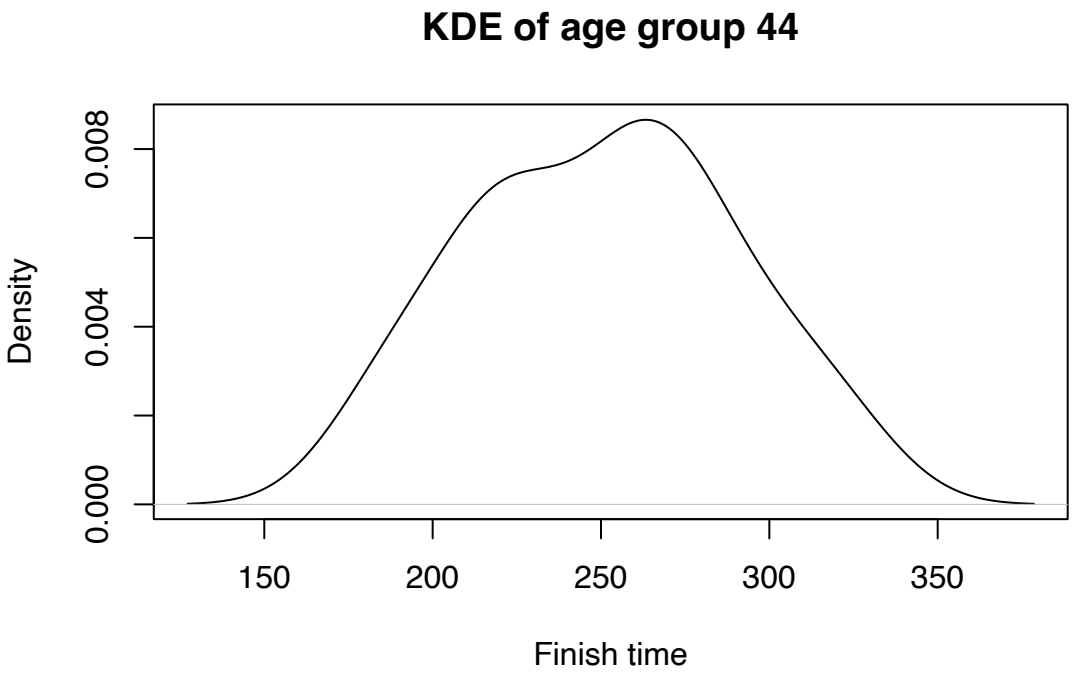
0.008
150
200
250
300
350
400
Finish time


KDE of age group 43
Density
0.000
0.004
100
150
200
250
300
350
400
Finish time


KDE of age group 44
Density
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0.004
0.008
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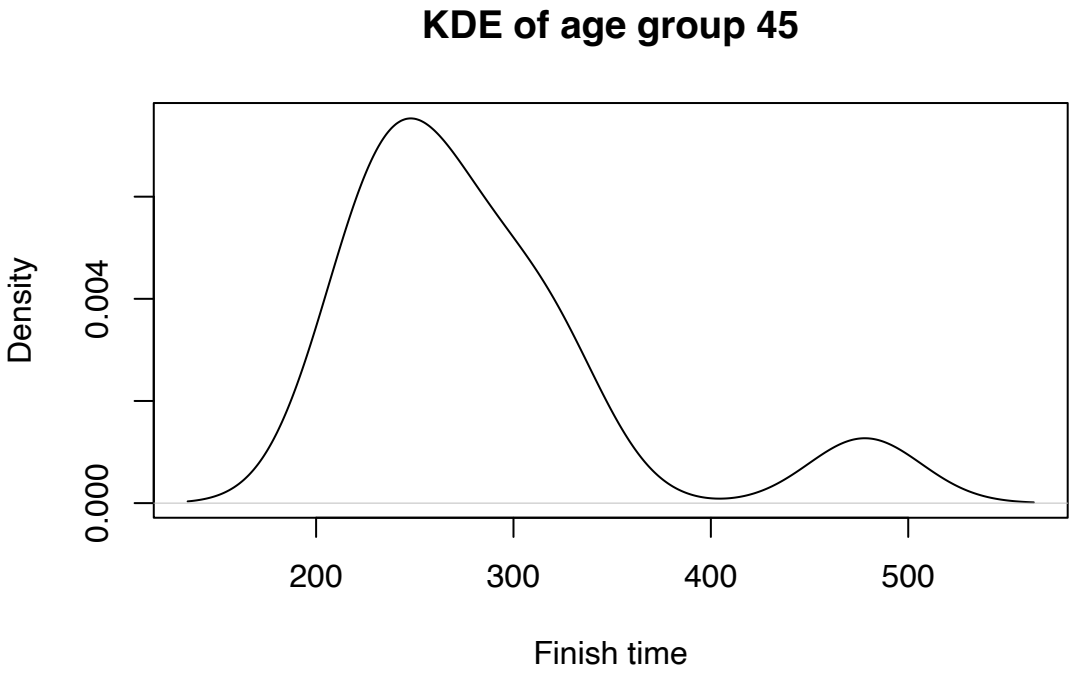
200
250
300
350
Finish time


KDE of age group 45
Density
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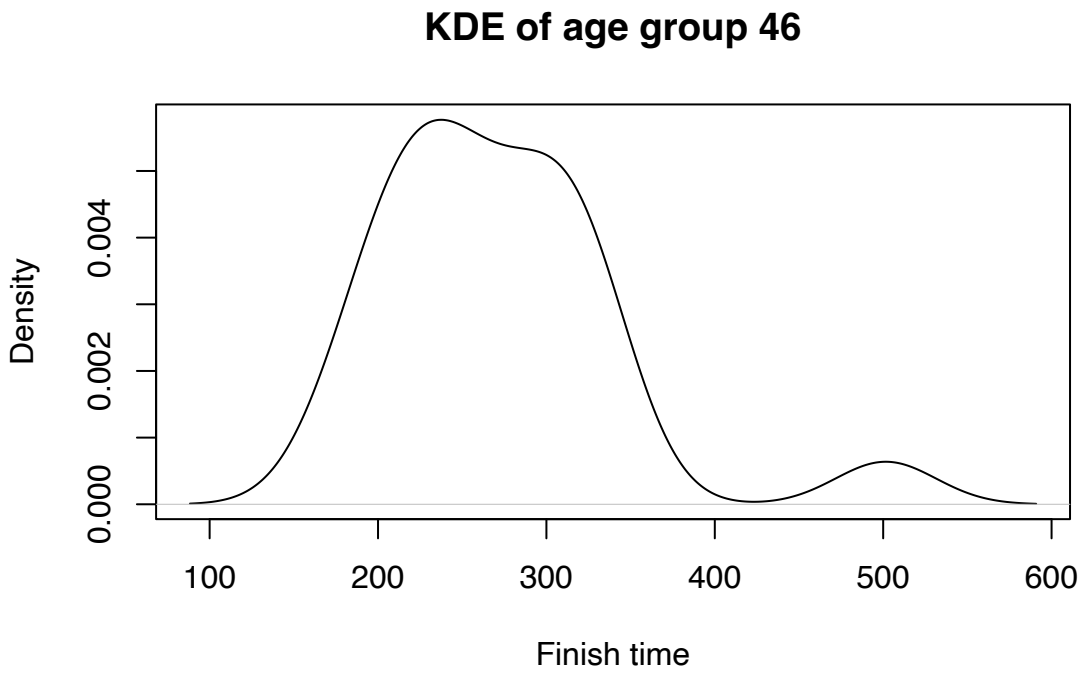
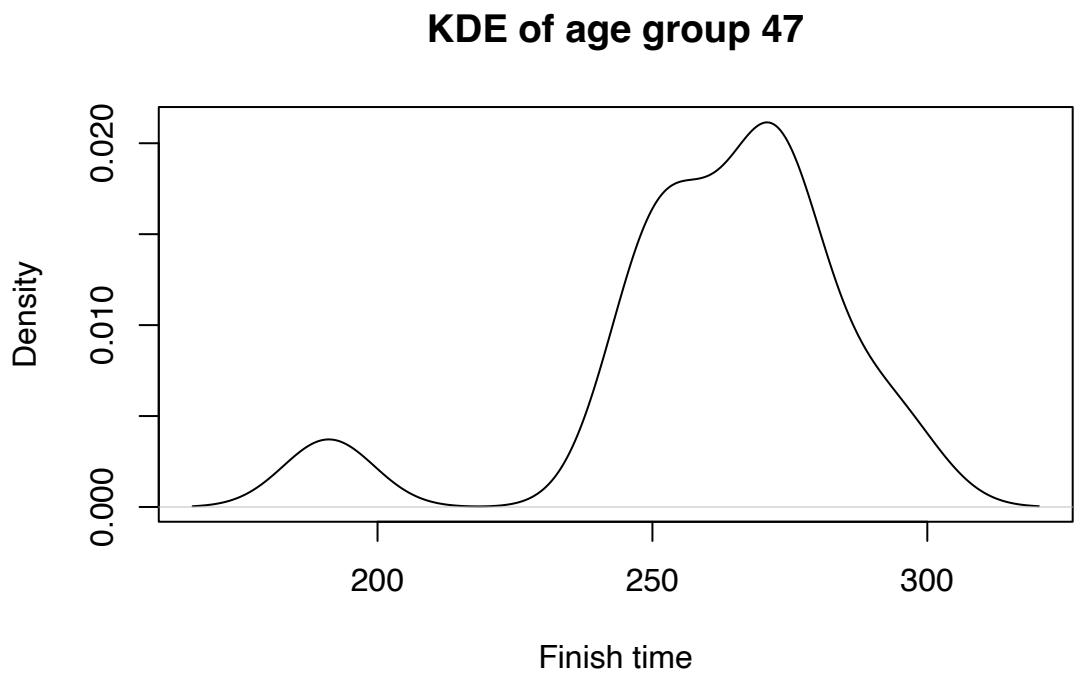
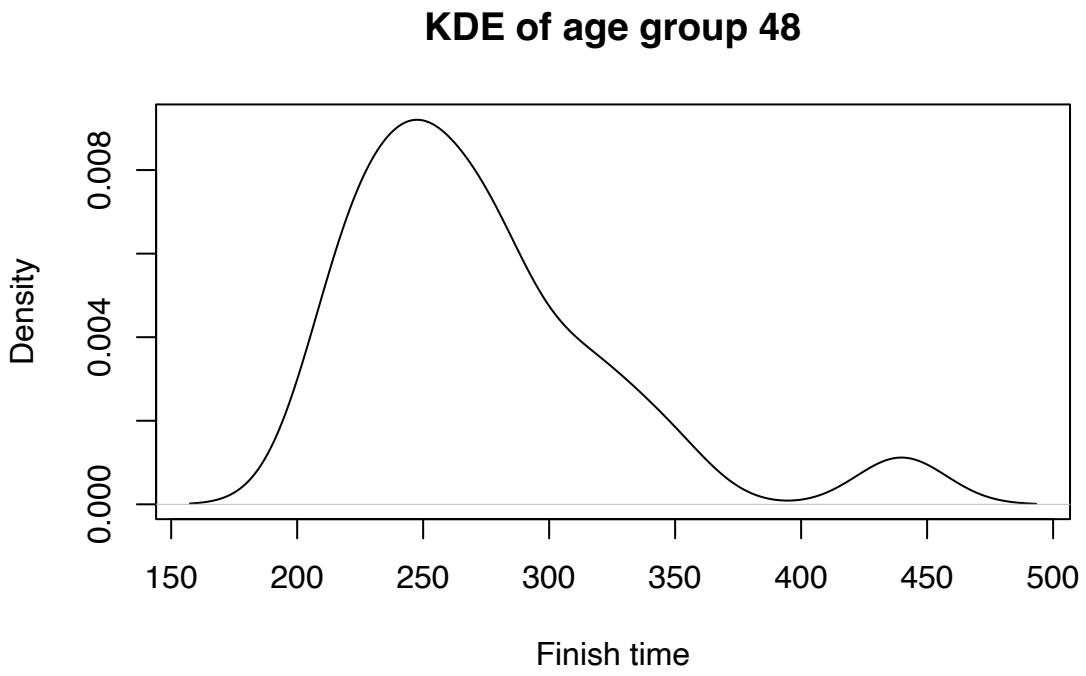
0.004
200
300
400
500
Finish time


KDE of age group 46
Density
0.000
0.002
0.004
100
200
300
400
500
600
Finish time


KDE of age group 47
Density
0.000
0.010
0.020
200
250
300
Finish time


KDE of age group 48
Density
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0.004
0.008
150
200
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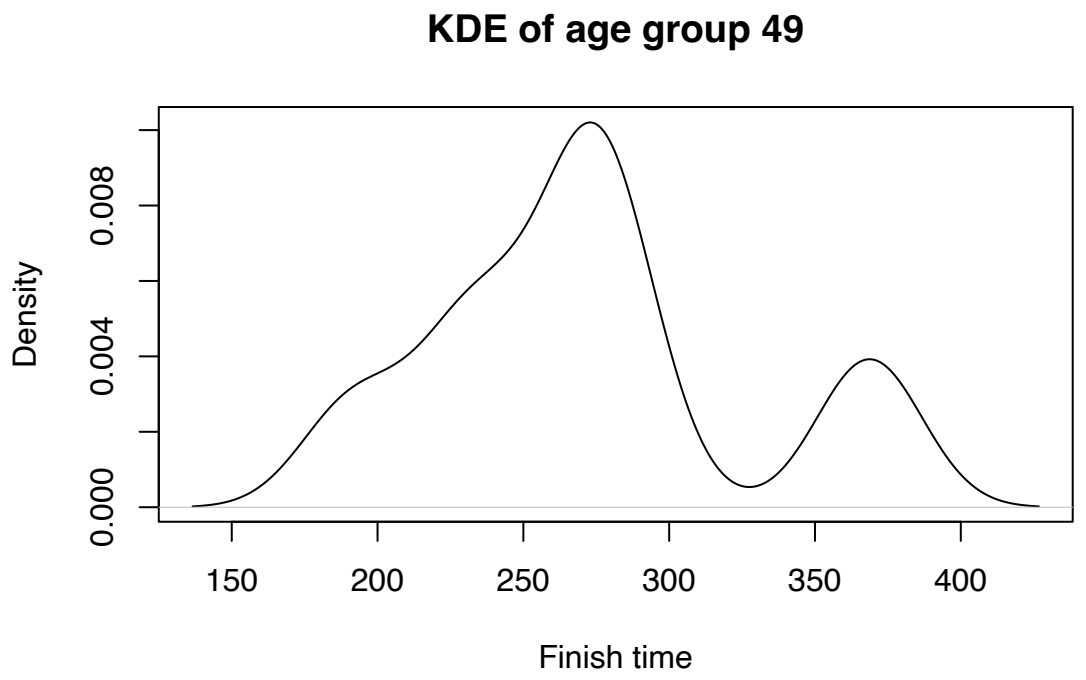
300
350
400
450
500
Finish time


KDE of age group 49
Density
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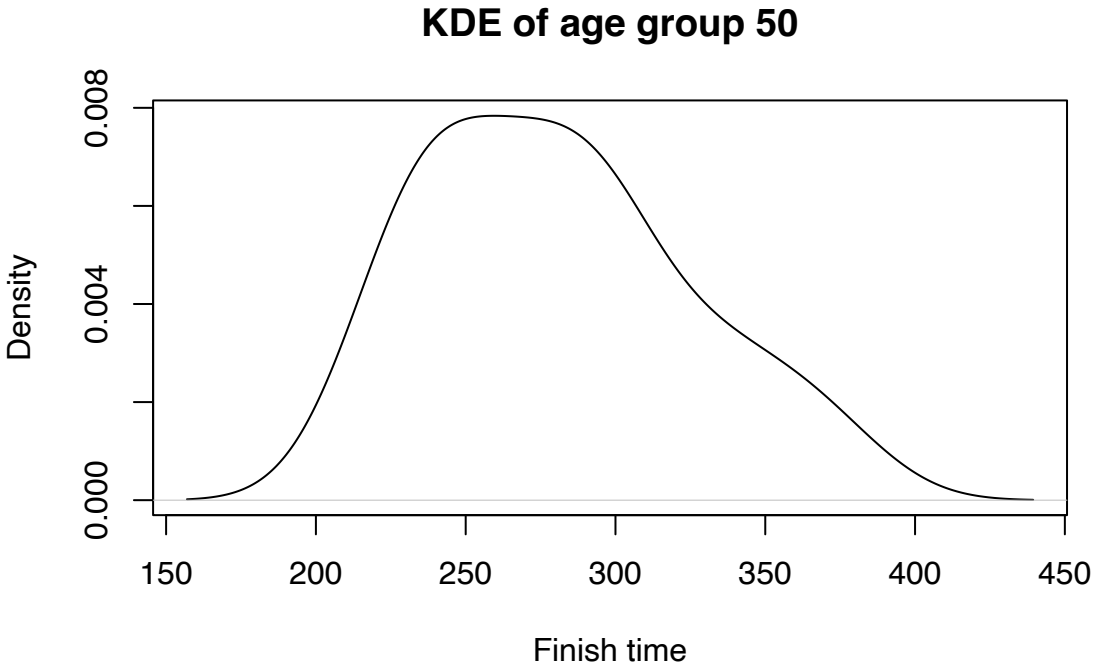
0.004
0.008
150
200
250
300
350
400
Finish time


KDE of age group 50
Density
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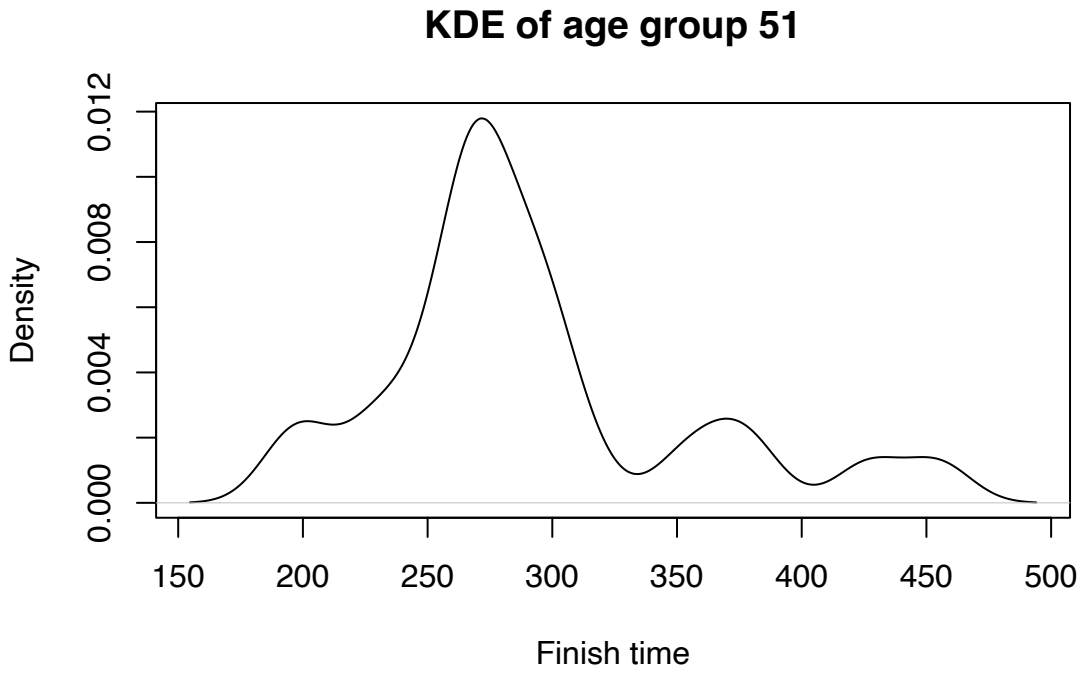
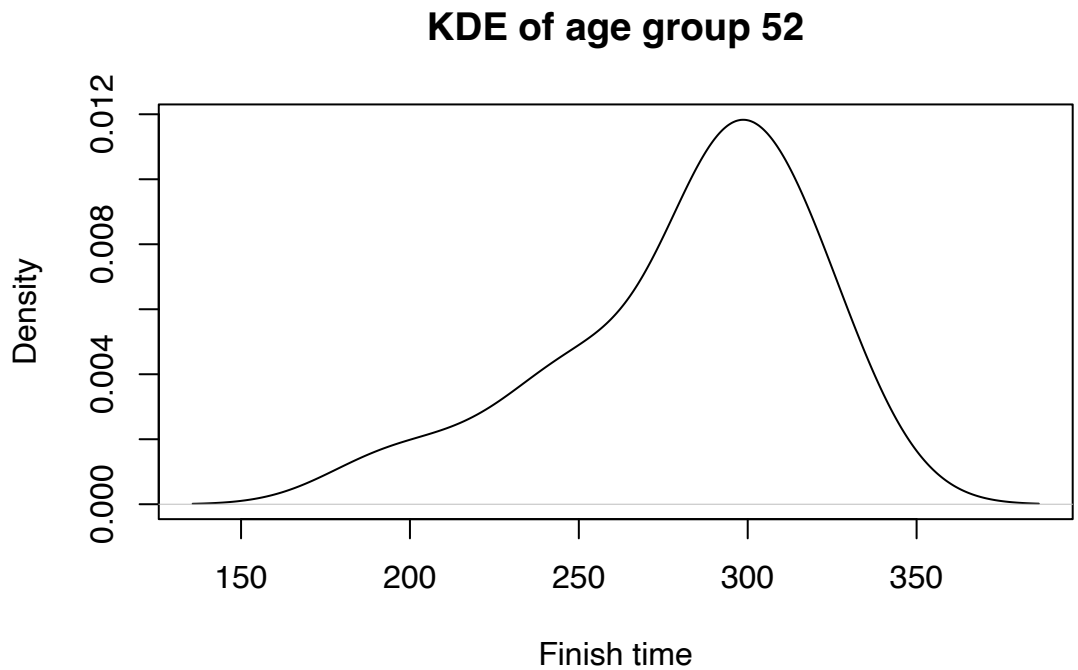
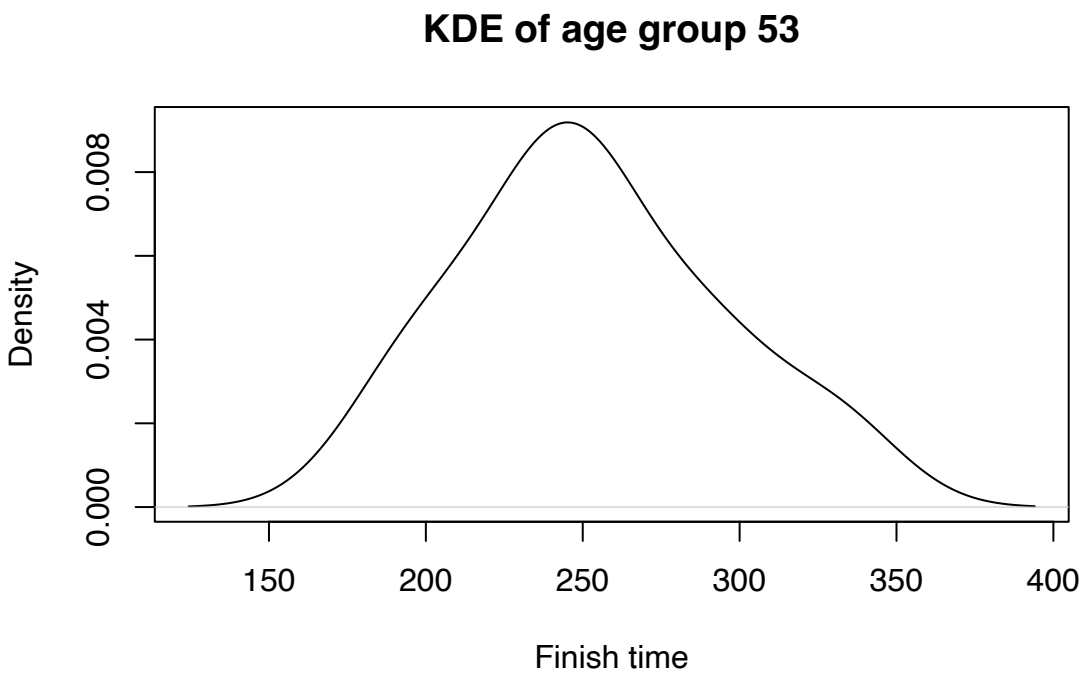
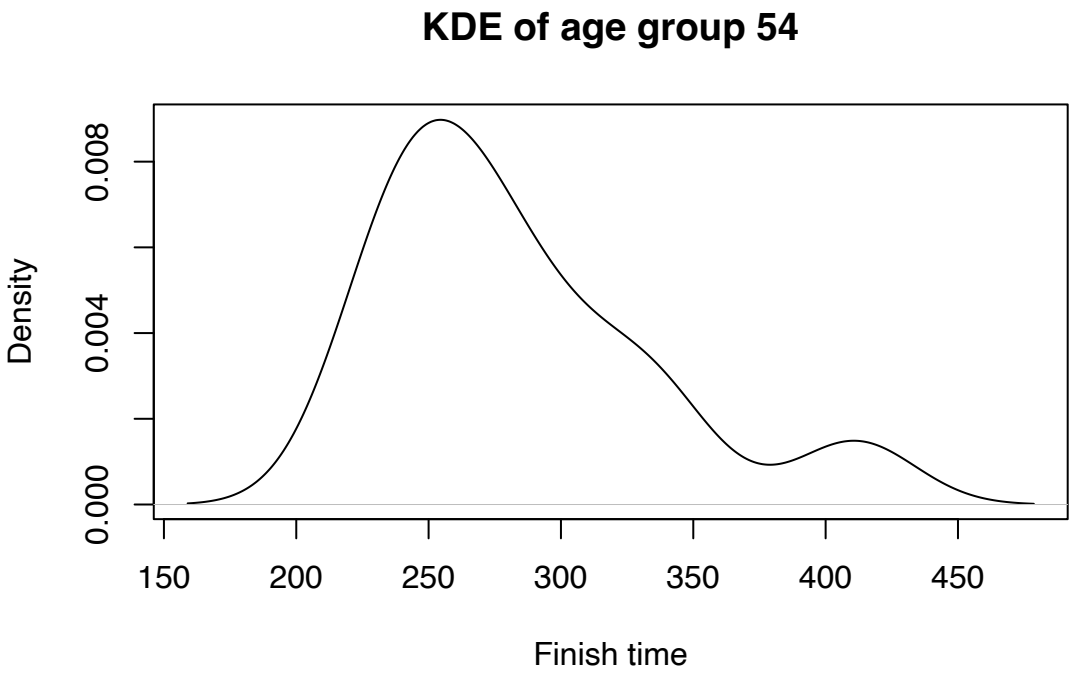
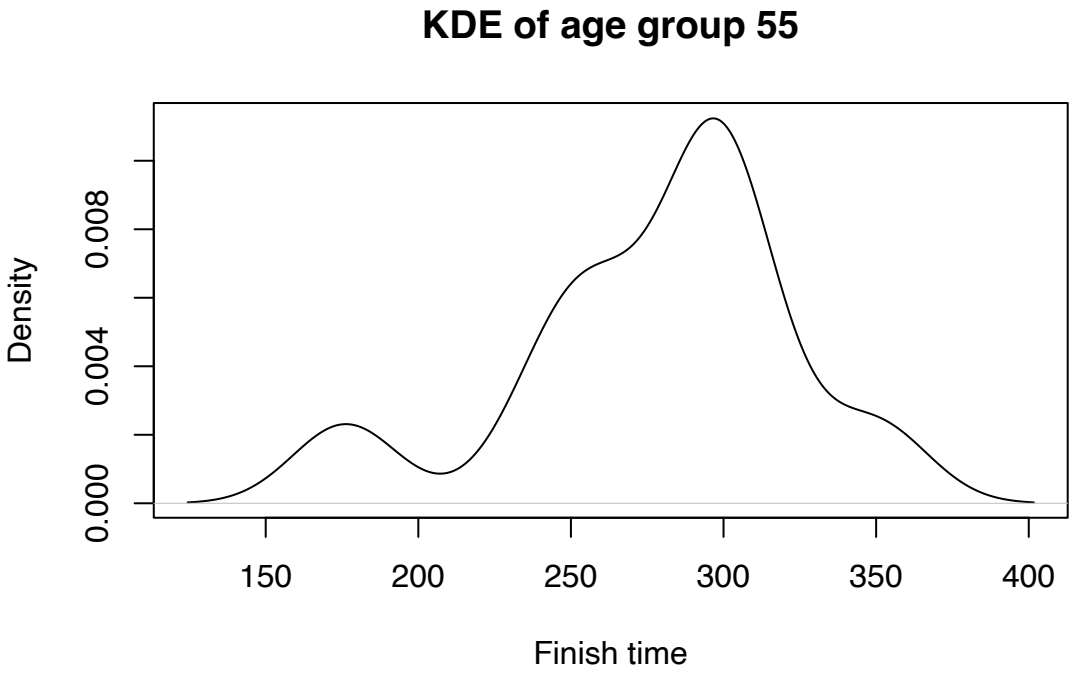
0.004
0.008
150
200
250
300
350
400
450
Finish time

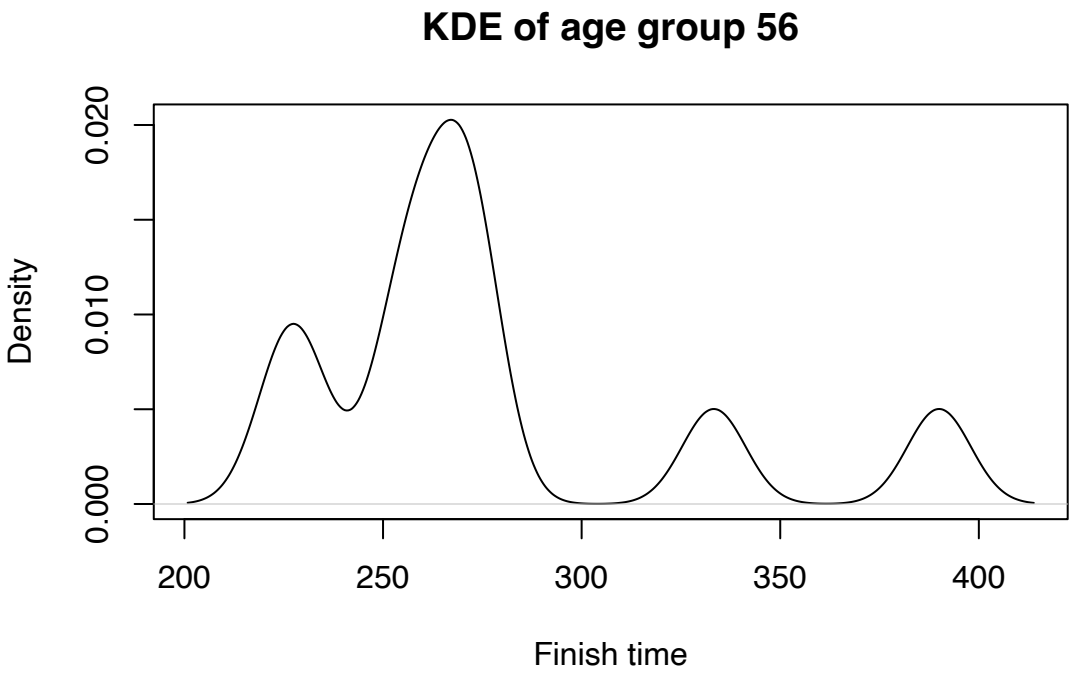

KDE of age group 51
Density
0.000
0.004
0.008
0.012
150
200
250
300
350
400
450
500
Finish time

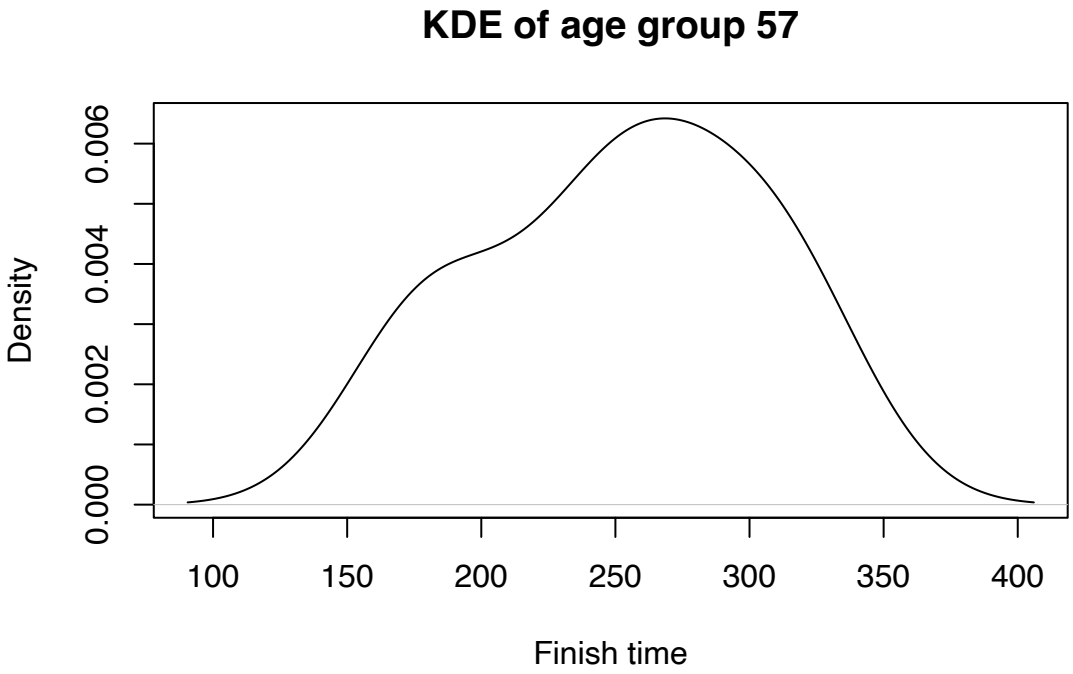

KDE of age group 52
Density
0.000
0.004
0.008
0.012
150
200
250
300
350
Finish time


KDE of age group 53
Density
0.000
0.004
0.008
150
200
250
300
350
400
Finish time


KDE of age group 54
Density
0.000
0.004
0.008
150
200
250
300
350
400
450
Finish time


KDE of age group 55
Density
0.000
0.004
0.008
150
200
250
300
350
400
Finish time


KDE of age group 56
Density
0.000
0.010
0.020
200
250
300
350
400
Finish time


KDE of age group 57
Density
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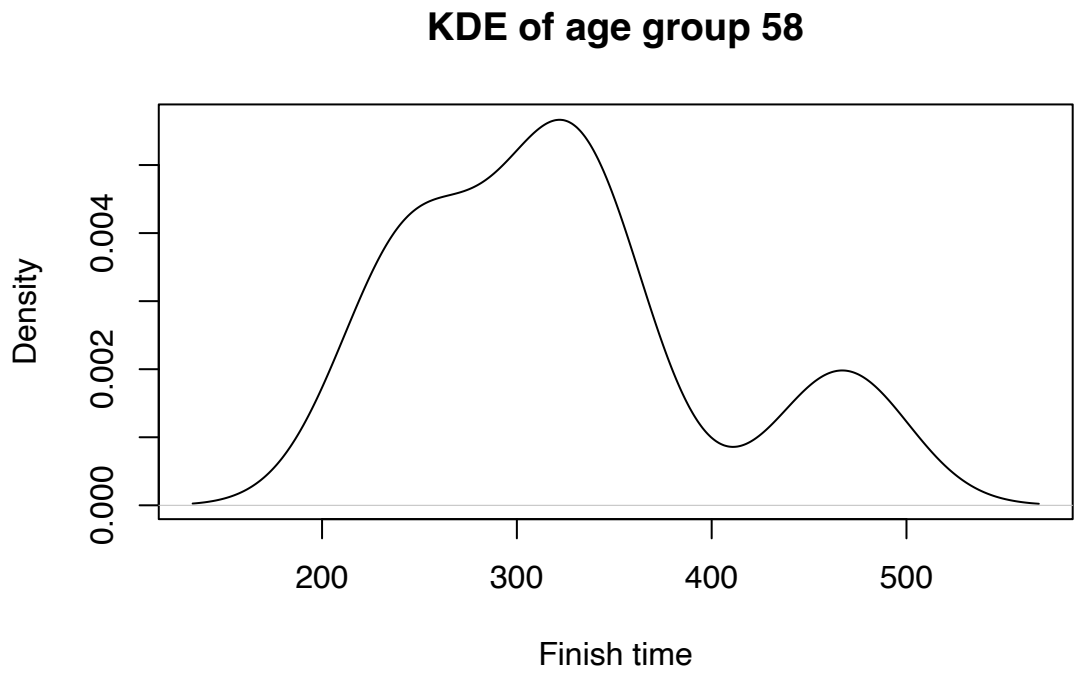
0.002
0.004
0.006
100
150
200
250
300
350
400
Finish time


KDE of age group 58
Density
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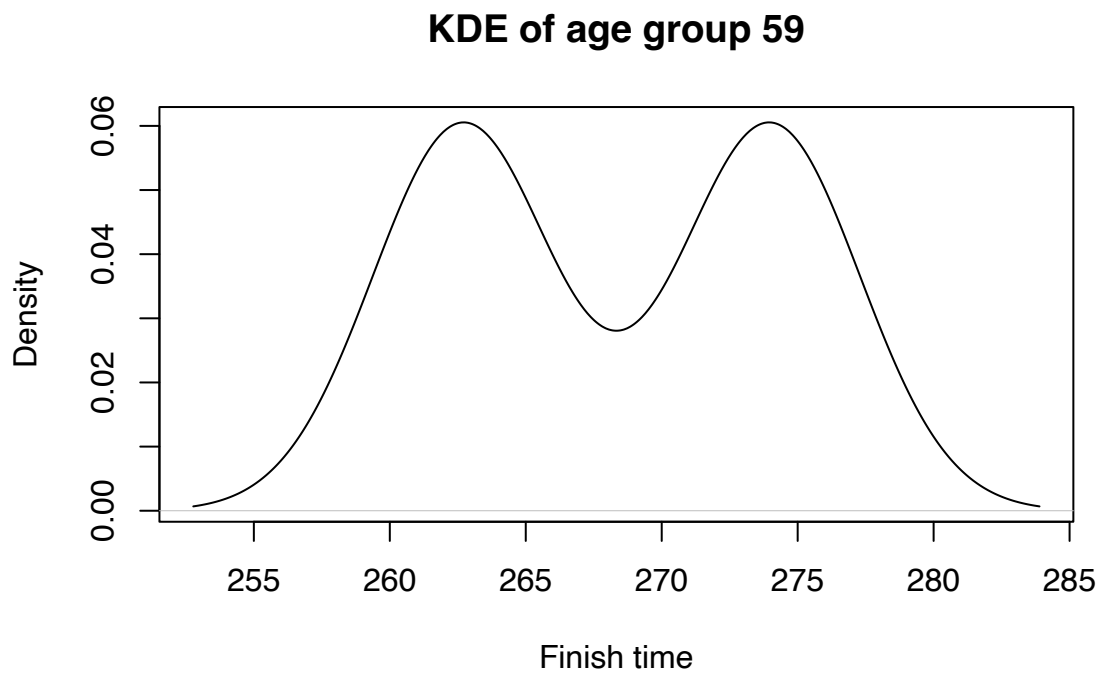
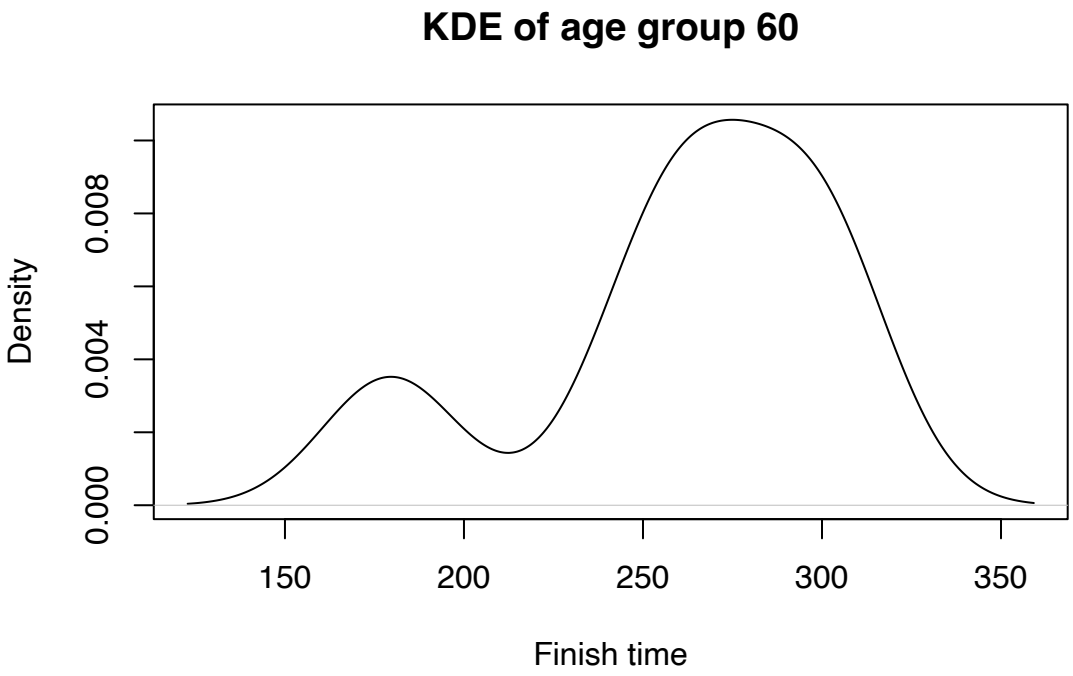
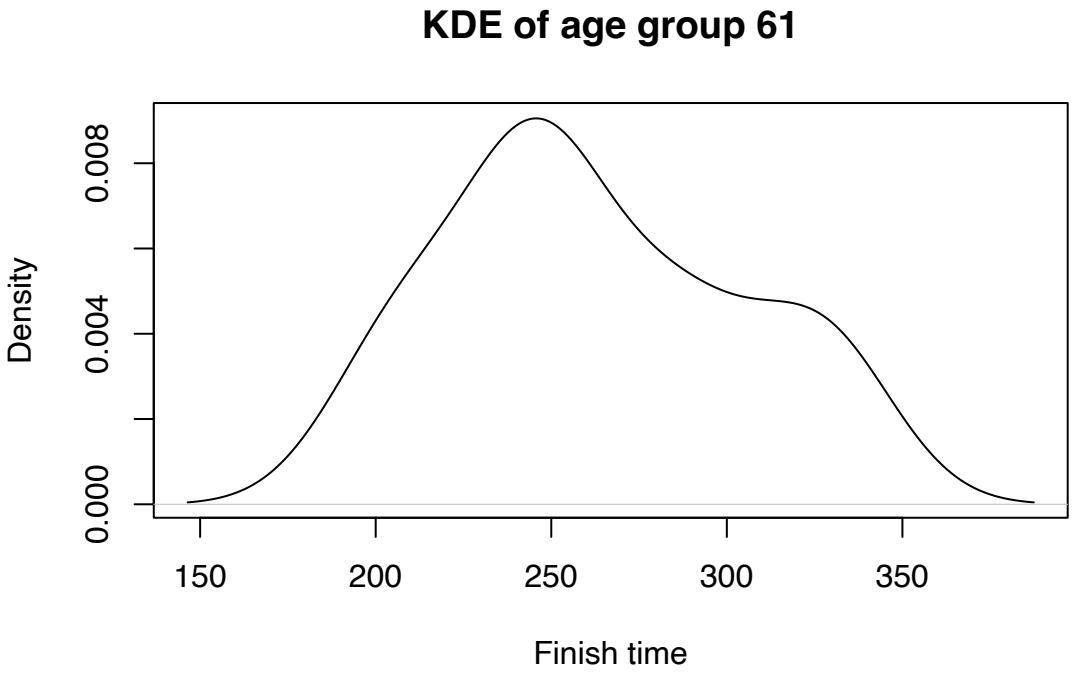
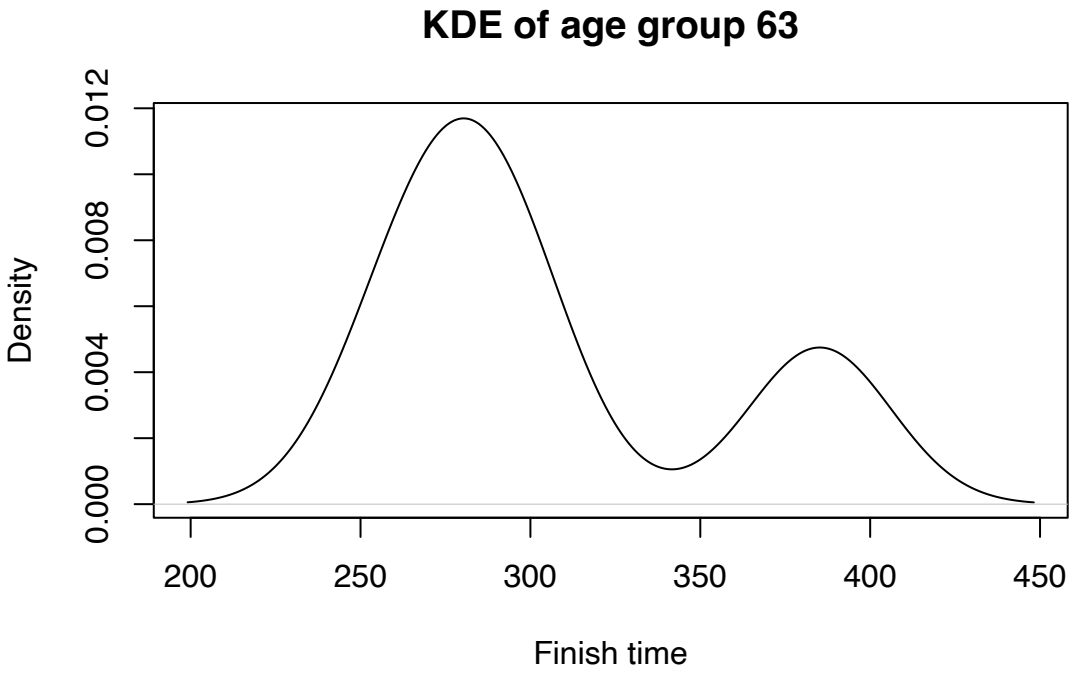
0.002
0.004
200
300
400
500
Finish time

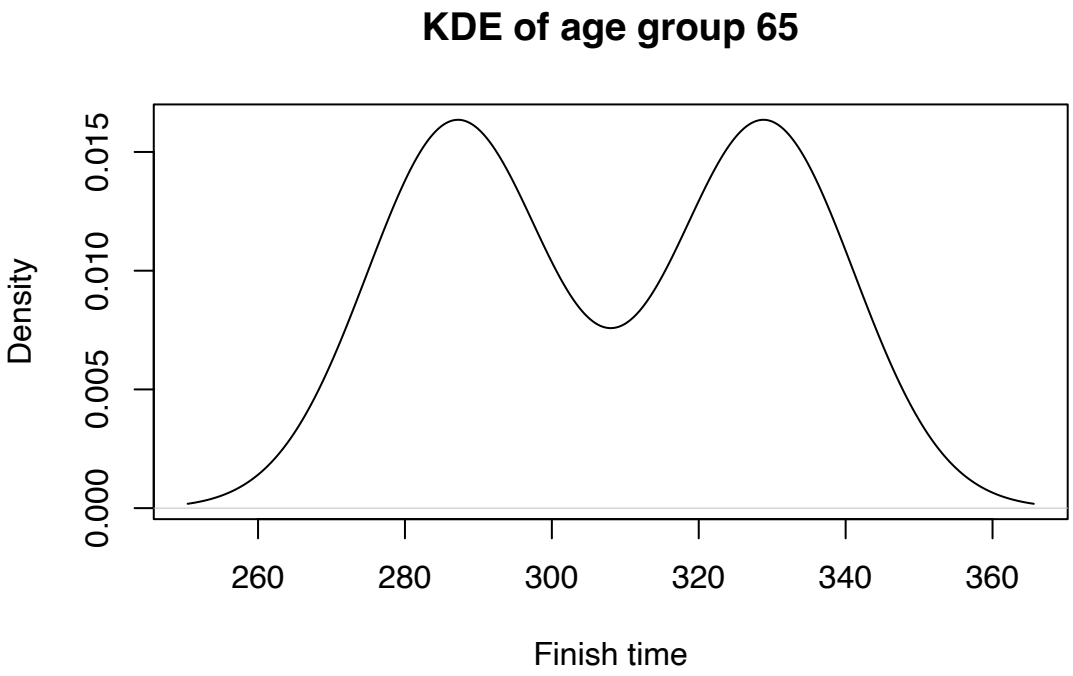

KDE of age group 59
Density
0.00
0.02
0.04
0.06
255
260
265
270
275
280
285
Finish time


KDE of age group 60
Density
0.000
0.004
0.008
150
200
250
300
350
Finish time


KDE of age group 61
Density
0.000
0.004
0.008
150
200
250
300
350
Finish time

KDE of age group 62
Density
0.000
0.002
0.004
200
300
400
500
Finish time


KDE of age group 63
Density
0.000
0.004
0.008
0.012
200
250
300
350
400
450
Finish time


KDE of age group 65
Density
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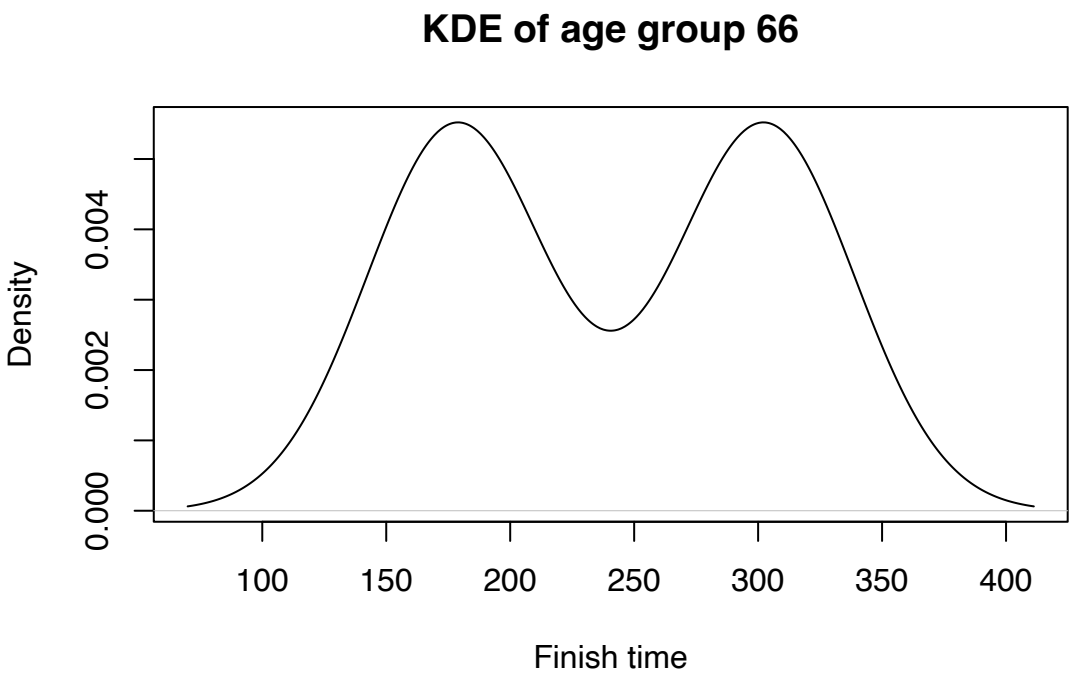
0.005
0.010
0.015
260
280
300
320
340
360
Finish time


KDE of age group 66
Density
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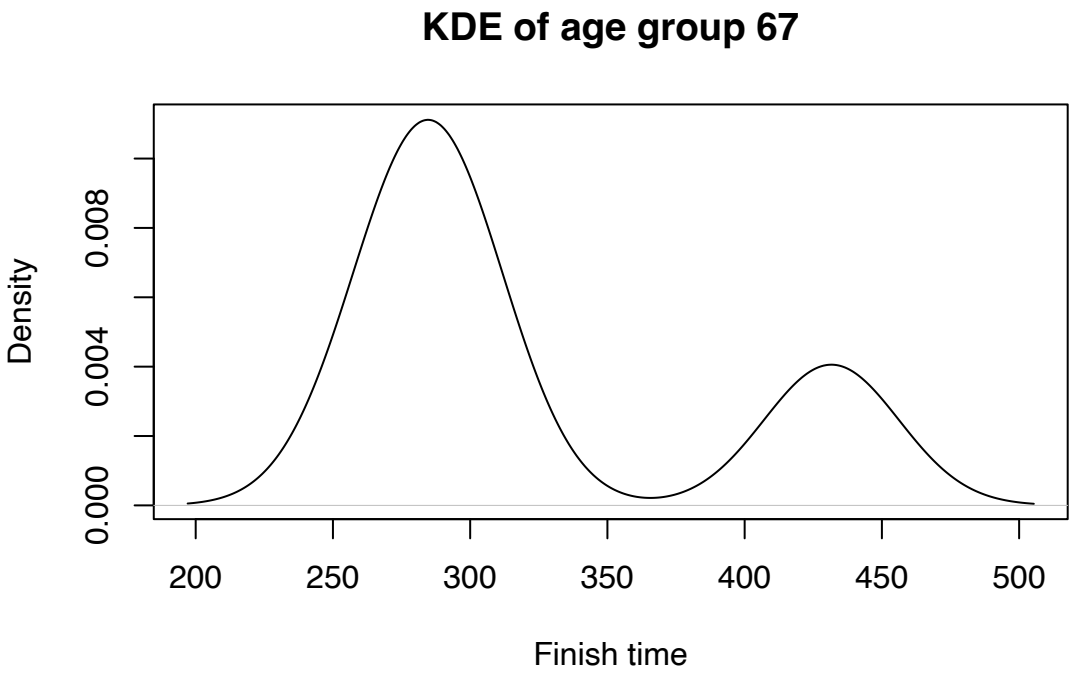
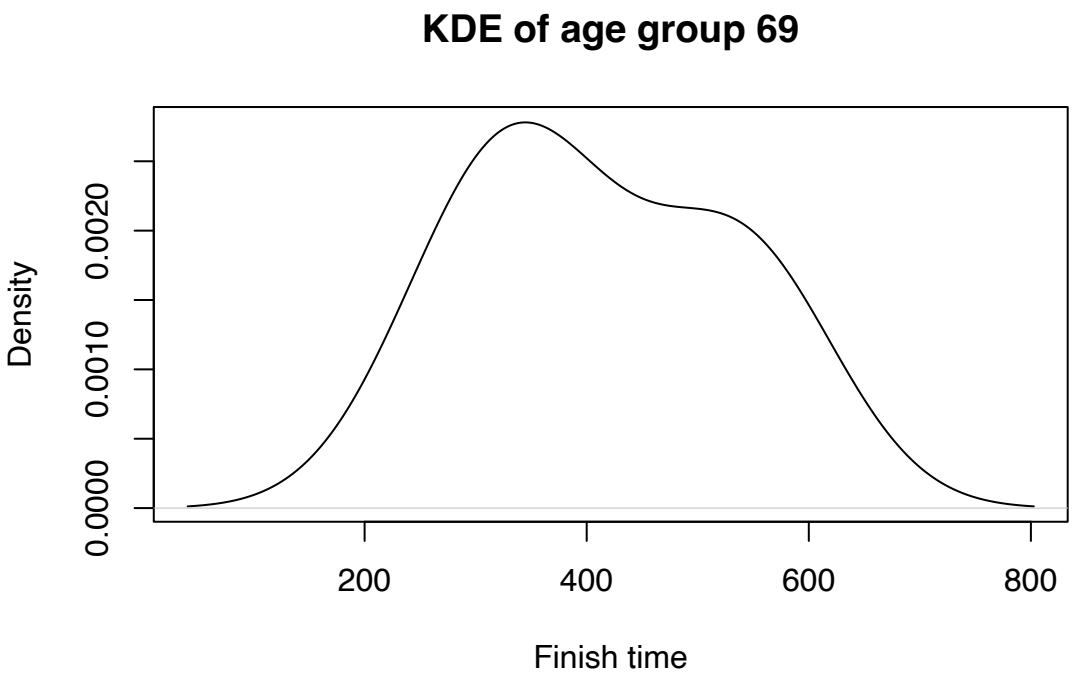
0.002
0.004
100
150
200
250
300
350
400
Finish time


KDE of age group 67
Density
0.000
0.004
0.008
200
250
300
350
400
450
500
Finish time


KDE of age group 69
Density
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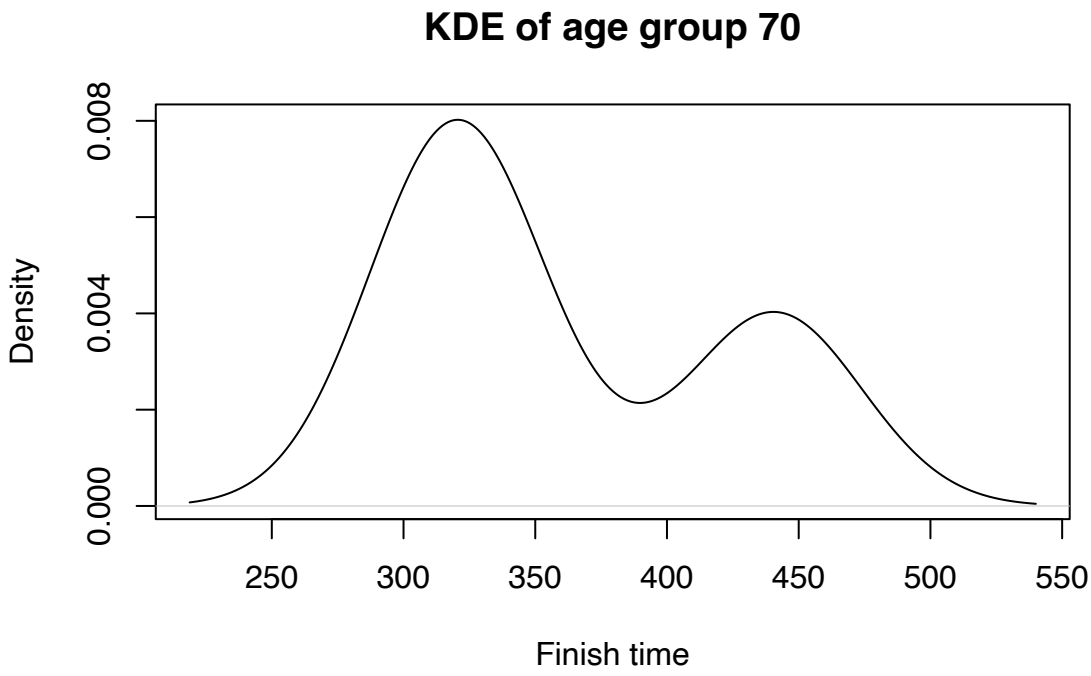
0.0010
0.0020
200
400
600
800
Finish time


KDE of age group 70
Density
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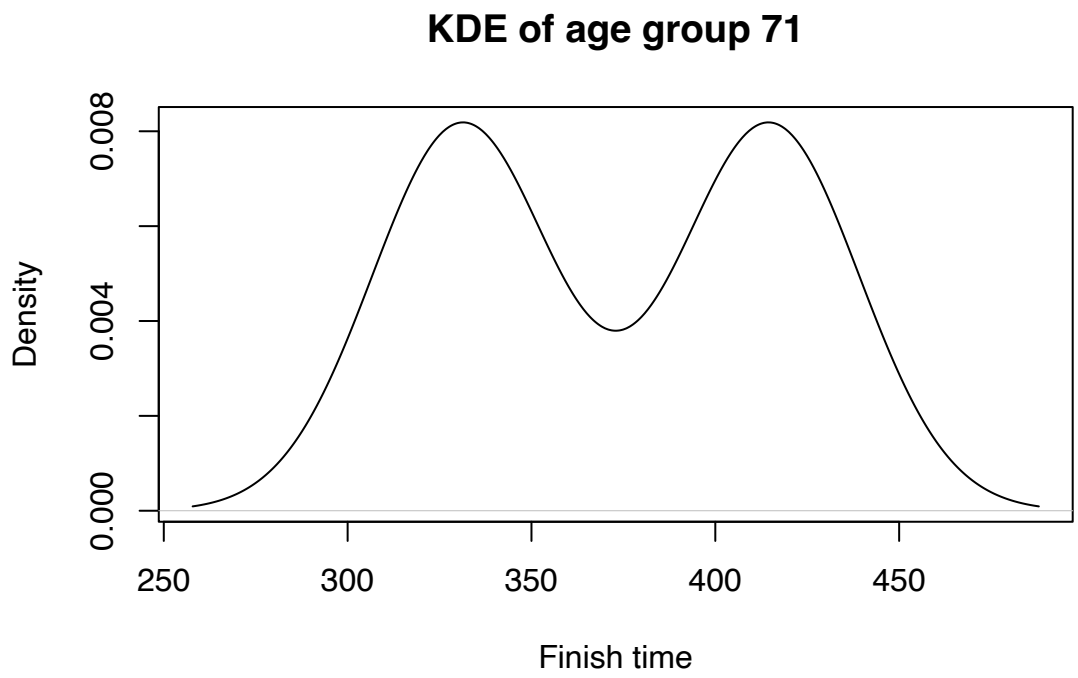
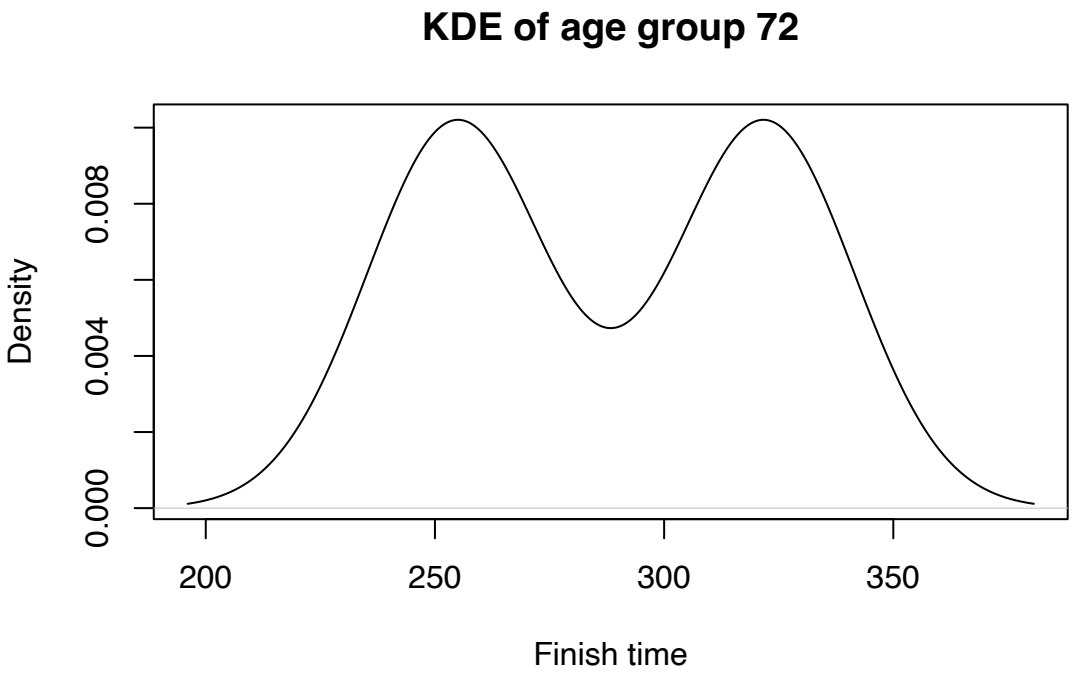
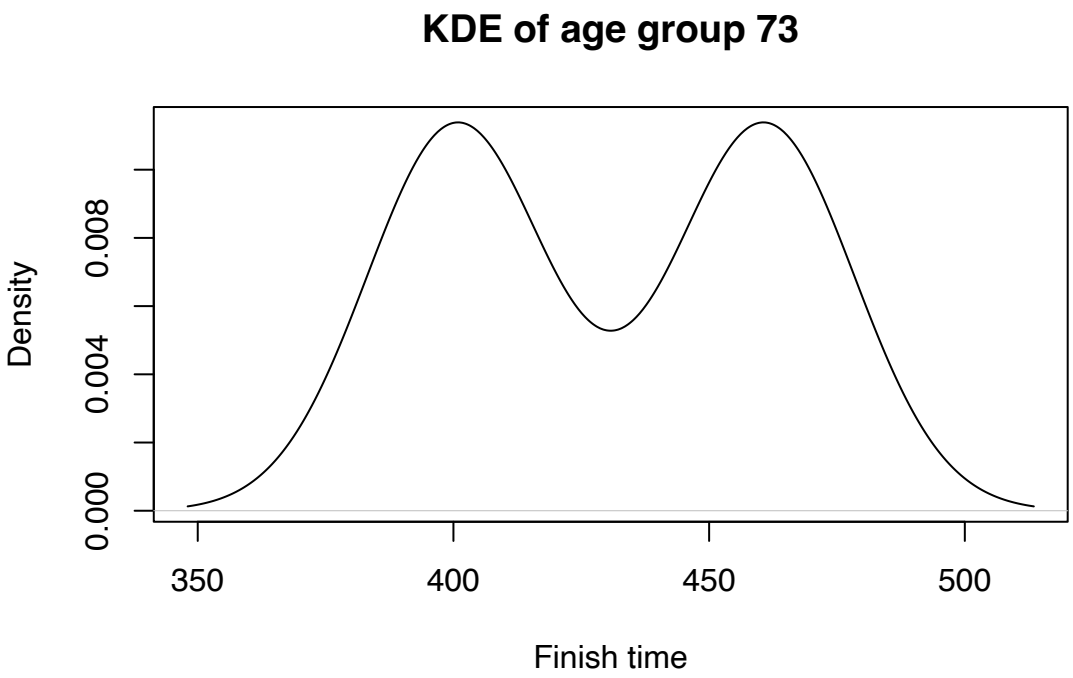
0.004
0.008
250
300
350
400
450
500
550
Finish time


KDE of age group 71
Density
0.000
0.004
0.008
250
300
350
400
450
Finish time





What do you observe in the plots? What might be a possible explanation? Is the median used in the previous point a good summary for each group?

Many groups have multiple modes, so the median might not be the best summary for every group. Though for some groups, it would be fine.